

The Transmission of Trade Shocks in Digital Markets*

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Abstract

We study how trade shocks are transmitted in digital marketplaces. Using product-level data from Amazon, we examine the 2025 escalation of U.S. tariffs on Chinese imports and the accompanying rise in trade policy uncertainty. We show that these shocks operate not only through prices, but also through non-price product attributes and the platform’s allocation of visibility. Retail pass-through is incomplete: sellers absorb part of the cost shock, while the post-tariff ranking environment becomes less favorable to costly products. A structural demand model decomposes the resulting change in consumer surplus into preference shifts, entry, exit, product attributes, and algorithmic allocation. Product turnover partly offsets losses from weaker attributes and reduced visibility, but the shock disproportionately displaces relatively expensive, lower-quality products.

JEL Classification Numbers: L81, E31, F13, F14, M31

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1 Introduction

How do trade shocks transmit when markets are intermediated by digital platforms? A large literature studies the incidence of tariffs through prices, asking how higher import costs are passed through to consumers and firms. In platform-mediated markets, however, competition depends on more than posted prices. Platforms determine which products consumers see, how prominently they are displayed, which sellers receive badges or fulfillment advantages, and how information about quality is presented. As a result, when trade costs change, adjustment may occur not only through prices, but also through product access, product attributes, platform allocation, and consumer demand. Standard pass-through frameworks therefore miss a central feature of digital markets: platforms mediate the transmission of trade shocks by allocating consumer attention which in turn affects sellers' incentive in pricing decisions and other decisions.

This paper studies how trade policy shocks affect outcomes on Amazon, a major digital marketplace. We focus on the sharp escalation of U.S. tariffs on Chinese imports in spring 2025, culminating in a 125 percent tariff rate on April 10, 2025. This tariff episode also generated a large increase in trade policy uncertainty, as policy changes were announced and implemented over a short period. Using high-frequency product-level data from Amazon search results, we examine how these shocks affect early-life product visibility, prices, non-price product attributes, search placement, product turnover, and welfare.

We document four transmission channels. First, heightened tariffs and trade policy uncertainty reduces the first-page representation of newly introduced Chinese products relative to comparable non-Chinese products. While these estimates do not directly identify entry, they show that newly introduced Chinese products become less likely to obtain prominent early-life visibility, consistent with a worsening of the cold-start problem under heightened uncertainty.

Second, the tariff increase leads to incomplete retail price pass-through and substantial adjustment along non-price margins. In more exposed product categories, we observe changes in Prime eligibility, platform-assigned badges, sales rank, sponsored placement, and the composition of brands and manufacturers shown to consumers. These results indicate that adjustment on digital platforms occurs through product attributes and paid visibility as well as through prices.

Third, the platform's ranking environment changes after the tariff shock. Because consumers typically engage with only the first page of search results, visibility on that page determines effective market access. We show that the mapping from product characteristics to search rankings on the first page shifts after the shock: products with less favorable demand-relevant characteristics become more likely to lose prominence. The platform therefore does not simply transmit the shock passively. It actively reallocates attention across products, amplifying the competitive consequences of higher trade costs.

Fourth, the structural analysis shows that the welfare consequences of these seller- and platform-side adjustments also depend on shifts in consumer demand. To quantify these effects, we estimate a structural demand model at the keyword-day level that allows price sensitivity and the valuation of product attributes to differ between the pre- and post-tariff periods and explicitly incorporates search position as a determinant of demand. Using instruments based on exchange rates, trade policy uncertainty, and differential tariff exposure, we recover economically meaningful demand elasticities and the relationship between visibility and demand. We then decompose changes in consumer surplus and seller profits into four components: demand shifts, product turnover, changes in product attributes, and changes in the platform’s allocation of visibility. The results show that cheaper product turnover partly offsets losses from weaker product attributes and a less favorable ranking environment among continuing products.

This paper contributes to two literatures. First, it extends the literature on tariff incidence by showing that in digital markets, trade shocks affect not only prices and quantities, but also product entry, product visibility, and seller composition, which together shape welfare. Second, it contributes to the literature on digital platforms by showing that platform design and ranking governance influence how macroeconomic shocks transmit through online markets. When platforms allocate consumer attention, the incidence and welfare effects of trade policy depend not only on pass-through, but also on how product characteristics are translated into exposure.

More broadly, the paper shows that understanding trade in the digital economy requires integrating cost shocks with platform-mediated competition. As a growing share of retail activity moves online, digital platforms increasingly shape how international trade policy affects consumers, sellers, and market structure. The remainder of the paper is organized as follows.

Section 2 reviews the related literature. Section 3 describes the data, Section 4 outlines the empirical setting, Section 5 presents the reduced-form evidence, Section 6 develops the empirical model and welfare analysis, and Section 7 concludes.

2 Related

Literature

This paper sits at the intersection of the literatures on tariff incidence and digital platform competition. The central insight of the paper is that, in platform-mediated markets, trade shocks transmit through more than retail prices. They also operate through product turnover, non-price product attributes, the platform’s allocation of visibility, and consumer preferences.

A large empirical literature studies how tariff shocks pass through into prices and how the incidence of those shocks is shared across firms and consumers. Recent work shows that tariff pass-through is heterogeneous across products and markets and need not map one-for-one from border prices into consumer prices. [Amiti et al. \(2019\)](#) find that the 2018 U.S. tariffs were largely

absorbed by domestic importers, while [Cavallo et al. \(2021\)](#) and [Cavallo et al. \(2025\)](#) document heterogeneous pass-through across product categories and persistent gaps between import and retail prices. A related literature emphasizes that trade shocks also trigger broader firm responses, including changes in sourcing, organization, and product mix. [Fajgelbaum and Khandelwal \(2022\)](#); [Fajgelbaum et al. \(2024\)](#); [Fajgelbaum and Khandelwal \(2024\)](#) document substantial reallocation in import patterns and supply chains during the U.S.–China trade war, while [Flaaen et al. \(2020\)](#) and [Hiller and Savage \(2021\)](#) show that tariff hikes affect domestic production, profitability, and employment. More generally, recent work highlights that trade shocks can propagate through production networks and induce persistent reconfiguration of supply chains, with implications for welfare and market structure ([Acemoglu and Tahbaz-Salehi, 2025](#); [Grossman et al., 2024](#)).

Our paper contributes to this literature by showing that, in digital retail markets, the transmission of trade shocks cannot be understood from price adjustment alone. Even when retail pass-through is incomplete, tariffs can still have large effects through changes in the set of products shown to consumers, in the observable attributes attached to those products, and in how the platform allocates visibility across listings.

A parallel literature in industrial organization and digital economics studies how platforms shape competition by designing the environment in which consumers and sellers interact. [Goldfarb and Tucker \(2019\)](#) argue that digital platforms act as market designers rather than neutral intermediaries, using ranking, recommendation, and information design to influence consumer behavior and seller competition. [Etro \(2023\)](#) shows that platforms adjust fees and commissions in response to sellers’ cost conditions, while [Teh \(2022\)](#) develops a framework in which platforms use non-price governance tools, including algorithmic visibility and entry restrictions, to mediate shocks. On the firm side, [De Loecker \(2011\)](#) emphasizes adjustment in product mix, which in a platform setting can take the form of withdrawal, relisting, or reallocation of seller effort across products. On the consumer side, [Fang et al. \(2024\)](#) show that buyer behavior is highly responsive to platform design and information presentation, and [Yu \(2024\)](#) shows that ranking and sponsored placement have important welfare consequences in online marketplaces.

Our paper contributes to this literature by showing that platform governance is itself a transmission mechanism for trade shocks. We show that the mapping from product characteristics to visibility changes after the tariff shock, so the platform’s ranking environment affects which products remain prominent and how the burden of the shock is distributed across sellers and consumers.

Taken together, the paper bridges these two literatures. The trade literature has primarily emphasized pass-through and reallocation, while the platform literature has emphasized visibility, information design, and governance. We combine these perspectives to show that in digital marketplaces trade shocks transmit through four related channels: product access and turnover,

prices and non-price product attributes, platform allocation of visibility, and consumer demand. This broader perspective is necessary to understand the incidence and welfare effects of trade policy in online retail markets.

3 Data

Amazon organizes products into roughly 40 broad categories, such as Electronics, Home & Kitchen, and Video Games, each of which contains many narrow subcategories. Our analysis focuses on Home & Kitchen, which contains more than 1,300 narrow subcategories, including Towels, Lids, and Robotic Vacuums. We restrict attention to the most detailed subcategories in Amazon’s category tree so that products within a keyword are as comparable as possible (see Figure A.1 in Appendix A.1). To further assess comparability within subcategories, we compute text-similarity measures based on product titles and descriptions. Figures A.2 and A.3 in Appendix A.1 report the distribution of these similarity scores and provide illustrative examples of products with low and high similarity.

We construct a product-level dataset of Amazon search results by using each narrow subcategory name as a search keyword. For each keyword, we scrape the first page of search results and record each product’s Amazon Standard Identification Number (ASIN), search position, and any platform-provided designations, including “Prime,” “Best Seller,” “Overall Pick,” “Sponsored,” and “Featured from Amazon Brands.”¹ For each listed product, we also collect price, average star rating, number of reviews, monthly sales, brand and manufacturer information, title and description, country of origin, and the date on which the product was first listed.² Panels A and B of Figure A.4 in Appendix A.1 show the interfaces from which these variables are collected. We repeat this scraping procedure almost daily over a period of slightly less than two months in order to capture time variation in search rankings and product attributes. We complement these cross-sectional snapshots with high-frequency historical data from the third-party provider Jungle Scout that records time-varying prices and sales ranks. Combining these sources allows us to observe both the cross-section of products shown in search results and within-product changes over time.

Table A.1 in Appendix A.2 reports summary statistics. The average product price is \$49.81.³ The

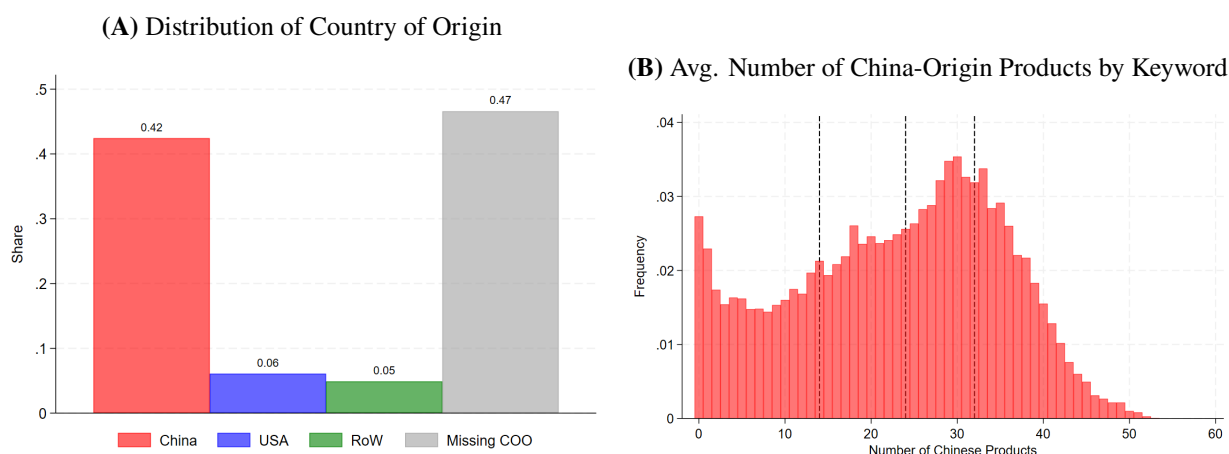
¹We do not directly observe Fulfillment by Amazon (FBA) status. We therefore use the Prime badge as a proxy, since Prime eligibility is closely tied to Amazon-handled fulfillment. This proxy is imperfect because some products may carry the Prime badge without being fulfilled through standard FBA arrangements.

²We restrict attention to the first page of search results because more than 70% of Amazon shoppers do not click beyond it (FTC, 2023). The first page typically displays about 60 products.

³Because the average price is well below the former \$800 *de minimis* threshold, many products in our sample would likely have qualified for duty-free entry if shipped individually to U.S. consumers. The termination of the *de minimis* exemption could therefore affect a substantial share of the products in our sample.

average sales rank is 160,996, and the average product sells 1,362 units in the previous month. Products in Home & Kitchen face an average referral fee of 14.81%. The average star rating is 4.46, the mean number of customer reviews is 3,370, and 73% of products carry the Prime badge. Panel A of Figure 1 reports the distribution of country of origin among products for which this information is available. Country-of-origin information is displayed for 88,020 of the 196,474 products in our sample.⁴ Country-of-origin information is missing for 47% of products; the remaining 53% are classified as originating from China (42%), the United States (6%), or the rest of the world (6%). These patterns indicate substantial exposure of Amazon search results to Chinese-origin goods, although the observed shares should be interpreted with caution because country-of-origin information is not available for all products. Appendix Table A.2 in Appendix A.4 reports the full country-level distribution. Panel B of Figure 1 reports, across keywords, the number of China-origin products appearing on the first page of search results. Exposure varies substantially across search queries. The 25th, 50th, and 75th percentiles of this distribution are 14, 24, and 32 China-origin products, respectively. In the analysis below, we use this cross-keyword variation to distinguish between relatively more exposed and less exposed product markets.

Figure 1. Country of Origin of Products in First Page Search Results



Notes: Panel A reports the share of products by country of origin among products on Amazon. The figure distinguishes products originating from China, the United States, the rest of the world (RoW), and products with missing country of origin (COO). Panel B reports the distribution of the average number of China-origin products appearing on the first page of search results. The vertical dashed lines indicate the 25th, 50th, and 75th percentiles of this distribution.

⁴For products that do not display country-of-origin information on the product page, we collect additional information using *Rufus*, Amazon’s internal large language model. For each ASIN with missing country-of-origin information, the program accesses the corresponding Amazon product page, submits the query “What is the country of origin of this product?”, and records the reported answer when available.

This section describes the policy event that structures the empirical analysis. The central event is the sharp escalation in U.S. tariffs on Chinese imports in spring 2025, which provides the main cost shock in our empirical design. The same episode also generated a sharp increase in trade policy uncertainty. We discuss each dimension of the policy environment and describe how its timing maps into the empirical analysis.

4.1 Tariffs

U.S. tariffs on Chinese imports were elevated but relatively stable throughout the early 2020s. In spring 2025, this stability ended with a rapid escalation in tariff rates. The largest increase occurred on April 10, 2025, when the United States raised tariffs on Chinese imports to 125%. Figure 2 summarizes the evolution of tariff rates over time; Appendix A.5 and Appendix A.6 provide additional institutional detail.

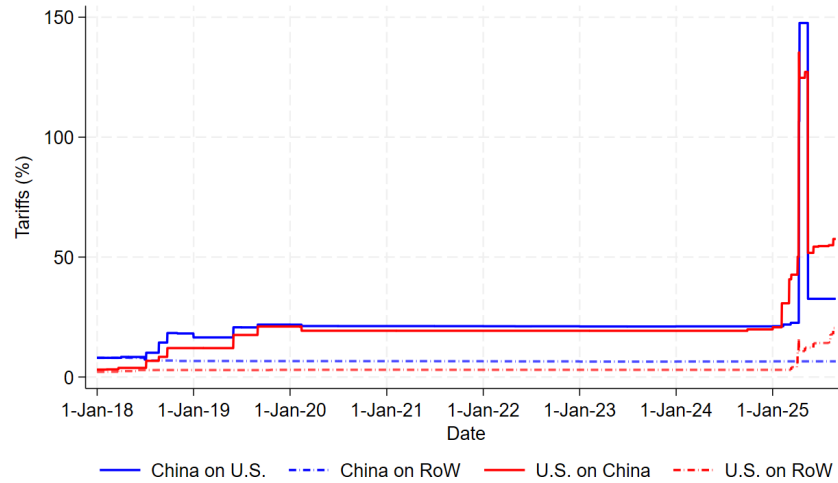
We use the sharp increase in tariffs as an exogenous trade shock. Product-level country-of-origin information allows us to classify products, and in turn search keywords, by their exposure to the shock. This variation identifies markets that were differentially affected by the tariff increase and provides the basis for estimating responses by consumers, sellers, and the platform.

A closely related policy change was the termination of the *de minimis* exemption for Chinese and Hong Kong-origin shipments. Before its removal, the exemption allowed individual shipments valued below \$800 to enter the United States duty-free and with limited customs processing, making it especially relevant for cross-border e-commerce sellers. On May 2, 2025, the United States revoked this exemption for Chinese and Hong Kong-origin shipments.⁵ We use this policy change as an additional robustness exercise, since it provides a closely related but distinct shock to the cost of low-value imports sold through digital marketplaces.

Following bilateral negotiations on May 11, 2025, the two governments reached a partial resolution. On May 14, 2025, the general tariff on Chinese imports was reduced from 125% to 10%. For shipments valued below \$800, the ad valorem rate was lowered from 120% to 54%, while the \$100 flat fee remained in place. We account for this sequence of policy changes in the timing of the empirical analysis.

⁵The policy was announced by executive order on April 2, initially imposing either a 30% tariff or a \$25 flat fee. These rates were increased to 90% (\$75) on April 8 and to 120% (\$100) on April 9.

Figure 2. Tariff Trends Timeline



Notes: This figure plots tariff rates between the U.S. and its trading partners—China and the rest of the world (RoW)—as well as between China and its trading partners—the U.S. and the RoW—over time. Data are from [Bown \(2025\)](#).

4.2 Trade

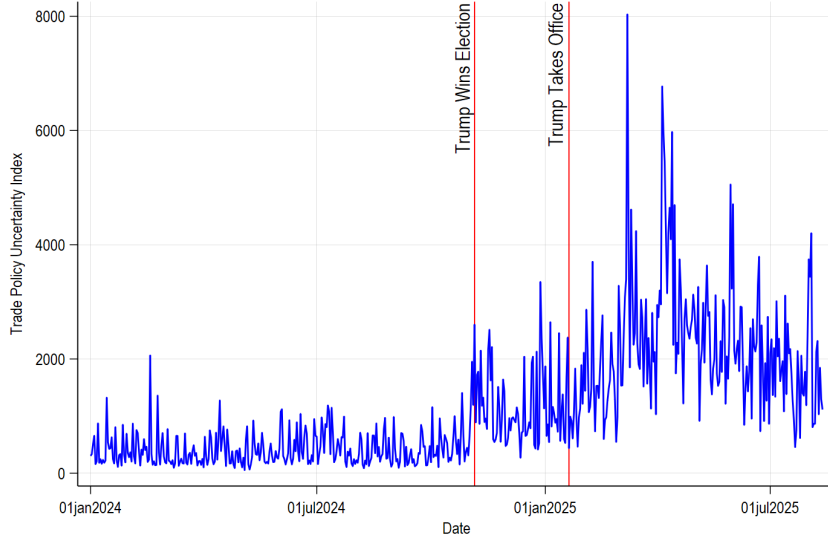
Uncertainty

Sudden and substantial changes in trade policy can heighten uncertainty for sellers and consumers, particularly when they are announced with limited advance notice. The tariff therefore have also generated a sharp rise in trade policy uncertainty. To capture this dimension of the policy environment, we use the U.S. Trade Policy Uncertainty Index constructed by [Baker et al. \(2016\)](#), which is based on the frequency of newspaper articles that jointly reference economic policy uncertainty and trade policy.

Figure 3 plots the daily evolution of this index. The series begins to rise after the November 2024 U.S. presidential election, consistent with increased attention to prospective tariff increases and broader changes in trade policy. The largest spike, however, occurs in April 2025, when tariffs between the U.S. and China escalated sharply. These movements indicate that the tariff episode substantially reduced the predictability of the trade environment, with the most pronounced increase in uncertainty coinciding with the sudden escalation in tariffs.

In addition to the direct cost effects of tariffs, trade policy uncertainty can affect both sides of the platform. On the demand side, uncertainty about future prices may change consumers' intertemporal purchasing decisions, leading them to accelerate or delay purchases. On the supply side, uncertainty about future costs and demand conditions may affect sellers' decisions about whether to introduce new products and how to price their products after entry.

Figure 3. U.S. Trade Policy Uncertainty Index



Notes: This figure plots the daily U.S. Trade Policy Uncertainty Index of [Baker et al. \(2016\)](#). The index rises with the frequency of articles in major U.S. newspapers that jointly reference economic policy uncertainty and trade policy.

5 Reduced-Form

Evidence

5.1 Product

Entry

We begin by studying product entry. Because we do not observe products that enter the marketplace but never appear on the first page of search results, we cannot estimate the effect of trade policy uncertainty on entry per se. We can, however, examine how the representation of newly introduced products among first-page search results evolves around the sharp increase in tariffs. This margin is particularly relevant in digital marketplaces, where new products must overcome a cold-start problem by first obtaining visibility before accumulating sales, reviews, and other signals that improve future placement.

We estimate separate event-study specifications for China-origin and non-China-origin products:

$$y_t = \alpha + \sum_{t \neq t^*} \beta_t \mathbb{1}\{\text{Week}_t = t\} + \varepsilon_t, \quad (5.1)$$

where y_t denotes the number of products that first become available in week t and appear on the first page of search results. The indicator $\mathbb{1}\{\text{Week}_t = t\}$ equals one in calendar week t , and t^* denotes the omitted reference week of the presidential inauguration.

The coefficients β_t measure changes in the first-page representation of newly introduced products relative to the pre-inauguration baseline. These estimates should therefore be interpreted as

changes in early-life visibility conditional on entry, rather than changes in entry itself.

The economic mechanism is straightforward. A less predictable trade environment may cause sellers to delay product launches or may make it more difficult for newly introduced products to secure early prominence in search results. In either case, heightened uncertainty can worsen the cold-start problem for exposed sellers.

5.1.1 Results

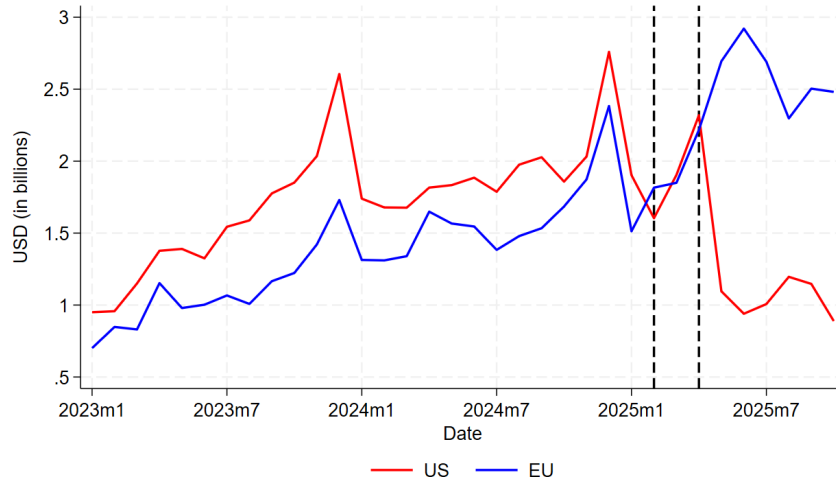
We begin by examining Chinese exports of low-value shipments, which are especially relevant for Amazon and especially exposed to the recent trade-policy changes, including the removal of the *de minimis* exemption. Figure 4 compares exports to the United States, which was directly affected by these policies, with exports to the European Union, which was not. The first vertical dashed line marks the presidential inauguration; the second marks the implementation of the tariffs.

Two patterns stand out. First, between the inauguration and the implementation of the tariffs,

Chinese exports to the United States remain elevated, consistent with some stockpiling in anticipation of higher trade costs. Second, once the trade measures take effect, exports to the

United States decline sharply, whereas exports to the European Union remain comparatively stable. Taken together, these patterns are consistent with heightened trade uncertainty followed by a sharp contraction in low-value shipments to the United States.

Figure 4. Chinese Low-Value Exports

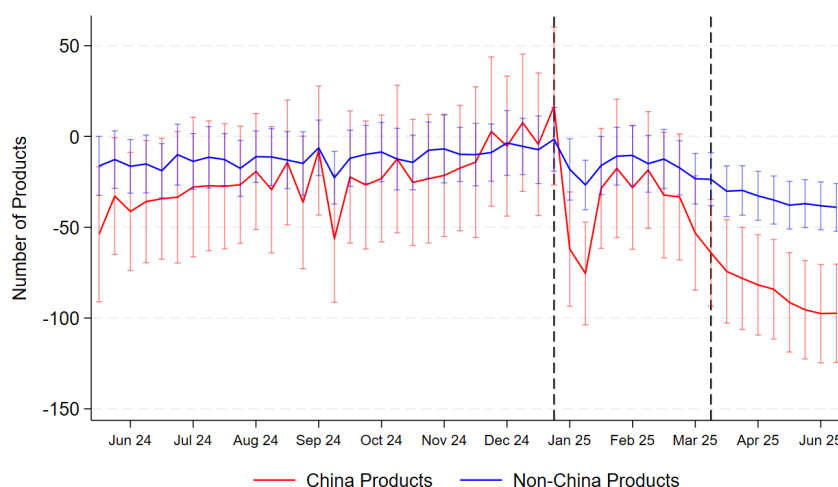


Notes: This figure plots monthly Chinese exports recorded under customs codes 9804, 9805, and 98010010 ([General Administration of Customs of the People's Republic of China, 2026](#)). The red series corresponds to exports to the United States, and the blue series corresponds to exports to the European Union. Export values are reported in billions of U.S. dollars. The first vertical dashed line marks the presidential inauguration, and the second marks the implementation of the tariffs.

We next turn to the representation of newly introduced products on the first page of Amazon search results. Figure 5 reports the event-study estimates from Equation (5.1) separately for China-origin and non-China-origin products. Before the inauguration, the two series move similarly, with little evidence of differential pre-trends. After the inauguration, however, they begin to diverge. Newly introduced China-origin products become less represented on the first page, while the corresponding series for non-China-origin products remains comparatively stable.⁶ This divergence becomes more pronounced after the April 10 tariff increase. The coefficients for China-origin products fall further and remain substantially below those for non-China-origin products. These estimates therefore suggest that the rise in trade policy uncertainty reduced the early-life visibility of newly introduced China-origin products and that the subsequent tariff increase intensified this relative decline.

Overall, the evidence indicates that the rise in tariffs and trade policy uncertainty is associated both with a contraction in low-value Chinese exports to the United States and with a deterioration in the early-life visibility of newly introduced China-origin products on Amazon. While these estimates do not identify entry directly, they show that newly introduced Chinese products became less likely to secure prominent early exposure, consistent with a worsening of the cold-start problem.

Figure 5. Product Entry



Notes: This figure plots event-study estimates of the number of newly introduced products appearing on the first page of Amazon search results. The red series corresponds to China-origin products, and the blue series corresponds to products originating outside China. Coefficients are normalized relative to the week preceding the presidential inauguration. Vertical bars indicate 95 percent confidence intervals. The first vertical dashed line marks the presidential inauguration, and the second marks the implementation of the tariff increase on April 10.

⁶The general decline after the inauguration reflects the fact that newly introduced products are less likely to appear on the first page of search results and therefore to be collected through our web scrapping.

5.2 Prices and Non-Price Product Attributes

We next examine the second transmission channel: adjustment in prices and non-price product attributes. To estimate the effect of tariffs on product outcomes, we use a difference-in-differences design that exploits the sharp increase in U.S. tariffs on Chinese imports on April 10, 2025. We construct subcategory-level panels by aggregating product-level outcomes for items appearing on the first page of search results. Identification comes from cross-sectional variation in tariff exposure across product subcategories, measured by their pre-policy exposure to China-origin products.

Our baseline specification is

$$y_{ct} = \alpha + \beta High-Exposure_{ct} + \gamma Post_t + \delta (High-Exposure_{ct} \times Post_t) + \kappa_c + \varepsilon_{ct}, \quad (5.2)$$

where $y_{ct} = \frac{1}{n} \sum_{i=1}^n y_{it}$ denotes the outcome of interest in subcategory c and period t .⁷ The indicator $High-Exposure_{ct}$ equals one if subcategory c in period t contains more than 14 China-origin products and zero otherwise.⁸ The variable $Post_t$ equals one after April 10, 2025.

The fixed effects κ_c absorb time-invariant differences across search keywords. This specification compares narrowly defined product markets that are similar in product type and demand conditions but differ in their exposure to Chinese imports. We also estimate product-level specifications with product fixed effects and category-level specifications using alternative measures of exposure. Because we do not observe product-specific tariff schedules, exposure is measured at the keyword or page level using the number or share of China-origin products displayed.⁹

The coefficient of interest is δ , which captures how outcomes change after the tariff increase in more exposed relative to less exposed product markets. This reduced-form effect reflects not only tariff pass-through into prices, but also adjustment in other observable product attributes and seller behavior on the platform.

5.2.1 Results

We begin by examining high-frequency price data by country of origin. We classify products as affected if their country of origin is China and unaffected otherwise, and normalize each series to its level on April 10, 2025. Figure 6 plots the resulting time series. Prices of non-China-origin

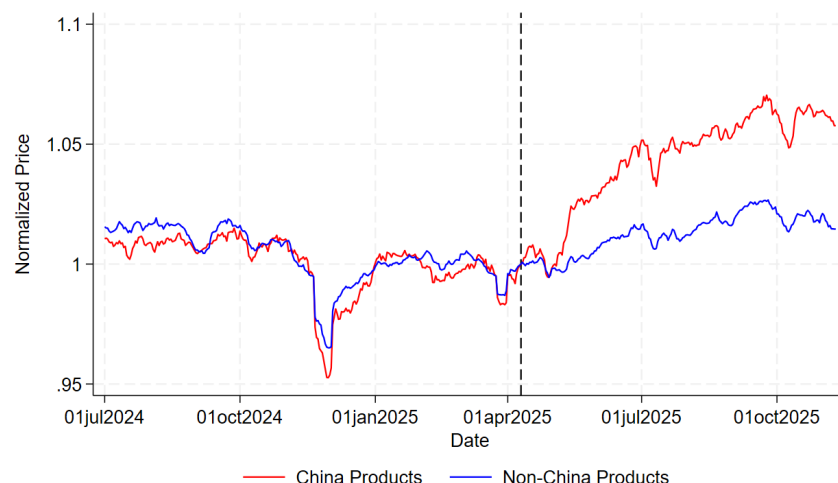
⁷Here y_{it} denotes the outcome for product i at time t , and n is the number of products observed in subcategory c .

⁸We group products originating in Hong Kong with products originating in China because the policy changes analyzed in this paper also applied to Hong Kong-origin products.

⁹A causal interpretation of the post-tariff estimates requires parallel pre-trends across more- and less-exposed markets. We assess this assumption using pre-shock event-time regressions; the results are reported in Table A.4 in Appendix A.8.

products rise modestly after the tariff announcement, while prices of China-origin products rise substantially more.

Figure 6. Price Time Series



Notes: This figure plots average normalized prices for products classified as affected (China-origin) and unaffected (non-China-origin) by the U.S. tariffs. Prices for each group are normalized to their level on April 10, 2025. The dashed vertical line marks the implementation date of the tariffs.

Table 1 reports the difference-in-differences estimates. Column (1) shows that prices rise by about 1.8 percent more in high-exposure subcategories than in less exposed subcategories after the tariff shock, consistent with incomplete retail pass-through. Adjustment, however, is not limited to prices. Prime eligibility declines in more exposed markets (Column 2), suggesting weaker fulfillment-related conditions after the shock. The share of products carrying the “Best Seller” badge also falls (Column 3), while there is no detectable change in the incidence of the “Overall Pick” badge (Column 4).

Sales performance worsens sharply in more exposed subcategories. Column (5) shows a significant deterioration in sales rank after the tariff, consistent with lower realized demand or weaker sales performance relative to less exposed markets. By contrast, average ratings and review counts change little in the short run (Columns 6 and 7).

The tariff shock also alters the composition of the displayed product set. Columns (8) and (9) show increases in both brand variety and manufacturer variety in high-exposure subcategories after the tariff. We interpret these changes not as an expansion in effective choice, but as evidence of greater turnover and relisting as sellers adjust their product mix. Column (10) shows that the number of sponsored products also increases in more exposed markets, consistent with greater reliance on paid visibility where the shock is strongest.

Taken together, these results show that tariff adjustment on digital platforms operates through

multiple observable margins. Prices rise, but so do changes in fulfillment status, badge assignment, sponsored placement, and the composition of products shown to consumers.

Table 1. Tariff Results

	Price	Prime Delivery	Best Seller	Overall Pick	Sales Rank	Ratings	Com- ments	Brand Variety	Manuf. Variety	# Sponsored
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>High-Exposure</i>	-0.004 (0.010)	0.064*** (0.013)	-0.007*** (0.002)	-0.004*** (0.001)	0.108*** (0.032)	-0.000 (0.001)	-0.059* (0.035)	0.202*** (0.027)	0.198*** (0.027)	0.141*** (0.021)
<i>Post</i>	-0.031*** (0.007)	-0.353*** (0.006)	0.002** (0.001)	-0.001 (0.000)	-0.031* (0.017)	-0.001 (0.001)	-0.357*** (0.020)	-0.016 (0.011)	-0.022** (0.011)	-0.013* (0.007)
<i>High-Exposure × Post</i>	0.018** (0.007)	-0.028*** (0.007)	-0.003*** (0.001)	0.000 (0.000)	0.054*** (0.018)	-0.001 (0.001)	-0.029 (0.023)	0.037*** (0.011)	0.040*** (0.011)	0.017** (0.007)
Keyword FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,782	21,782	21,782	21,782	21,782	21,782	21,782	21,782	21,782	21,782

Notes: This table shows the estimates from Equation 5.2. Column (1) shows results related to price. Columns (2)–(10) show results related to delivery, platform-assigned badges, sales, ratings, comments, brand variety, manufacturer variety, and the number of sponsored products. Robust standard errors are clustered at the keyword level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendices A.9 to A.11 report a range of robustness checks, including alternative exposure cutoffs, alternative shock timing, and alternative levels of aggregation. The results remain qualitatively similar across these specifications.

5.3 Platform Allocation of Visibility

The third transmission channel operates through the platform’s allocation of visibility. Because the platform’s exact ranking rule is unobserved, we approximate the relationship between product placement and observable product characteristics using the following reduced-form specification:

$$r_{jct} = \mathbf{X}'_{jct}\beta + \zeta_j + \kappa_c + \tau_t + \epsilon_{jct}, \quad (5.3)$$

where $\mathbf{X}_{jct}$ includes product attributes such as price, badges, ratings, and sales rank; ζ_j captures time-invariant product quality; κ_c captures keyword-specific differences across search markets; and τ_t captures common time effects. The residual ϵ_{jct} represents idiosyncratic variation in placement. Appendix A.12 provides descriptive evidence for the inclusion of these variables.

To study how the platform’s ranking environment changes after the tariff shock, we estimate Equation (5.3) separately in the pre- and post-tariff periods. Let β^0 and β^1 denote the estimated coefficients before and after the tariff, respectively. We then use these estimates to predict each product’s placement under four combinations of attributes and ranking environments: pre-tariff attributes under the pre-tariff ranking environment, post-tariff attributes under the pre-tariff

ranking environment, pre-tariff attributes under the post-tariff ranking environment, and post-tariff attributes under the post-tariff ranking environment. These predictions allow us to decompose the total change in placement into an attribute component and a ranking component:

$$\underbrace{\hat{r}^{1,1} - \hat{r}^{0,0}}_{\text{Total Change}} = \underbrace{(\hat{r}^{0,1} - \hat{r}^{0,0})}_{\text{Attribute Change}} + \underbrace{(\hat{r}^{1,1} - \hat{r}^{0,1})}_{\text{Ranking Change}}. \quad (5.4)$$

The first term captures how changes in observable product characteristics would have affected placement if the ranking environment had remained fixed. The second captures how the mapping from characteristics to placement changed over time, holding observed characteristics fixed. A positive value indicates worse placement, that is, movement to a lower position on the page.

5.3.1 Results

Table 2 reports the relationship between product placement and observable characteristics before and after the tariff increase. These coefficients can be interpreted as reduced-form weights relating product characteristics to visibility. The specification explains a large share of the variation in product placement, with R^2 values between 0.76 and 0.78 across periods. The coefficient on price rises from 4.5 to 5.9, implying that a given increase in price is associated with a larger deterioration in placement after the tariff shock than before it. In that sense, the post-tariff ranking environment becomes less favorable to higher-priced products, reinforcing the incomplete-pass-through pattern documented above. This interpretation is consistent with contemporaneous statements by Amazon's CEO, who stated on April 10, 2025, that the company would "try and do everything we can to keep prices as low as possible for customers" (Palmer, 2025; Linnehan, 2025).

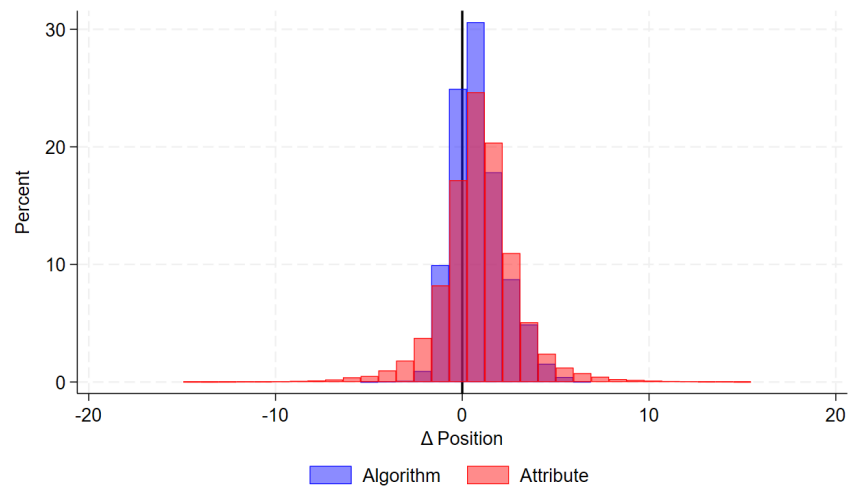
Figure 7 plots the distributions of the attribute-driven and ranking-driven components of the change in placement. The attribute component is shifted slightly to the right of zero, indicating that observed changes in product characteristics alone would have worsened placement after the tariff shock. The ranking component is also shifted to the right, implying that the post-tariff ranking environment further lowered visibility for affected products even holding attributes fixed. The total effect therefore reflects both weaker product characteristics and a less favorable mapping from those characteristics into visibility.

Taken together, these results show that the platform is not merely a passive conduit for the tariff shock. Changes in the ranking environment constitute a distinct transmission channel through which trade-policy shocks affect which products remain visible to consumers. Appendix A.12.3 reports two additional exercises, including a pre-period placebo test and an analogous analysis around the mid-May 2025 tariff reduction.

Table 2. Algorithmic Sensitivity of Product Visibility

	Price (1)	Prime (2)	Best Seller (3)	Overall Pick (4)	Rating (5)	Sales Rank (6)
<i>Pre-Tariff Position</i>	4.503*** (0.283)	-1.497*** (0.130)	0.576** (0.250)	-9.522*** (0.308)	-1.772*** (0.339)	4.071*** (0.097)
<i>Post-Tariff Position</i>	5.933*** (0.400)	-0.823*** (0.112)	0.574* (0.301)	-10.494*** (0.348)	-1.246*** (0.399)	4.100*** (0.113)
Keyword FE	Yes	Yes	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	877,903	877,903	877,903	877,903	877,903	877,903

Notes: This table reports estimates of the relationship between product position and product attributes before and after the implementation of tariffs. Standard errors are clustered at the keyword–ASIN level and shown in parentheses. The pre-tariff sample contains 535,596 observations, and the post-tariff sample contains 342,307 observations. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Figure 7. Changes in Product Placement

Notes: This figure decomposes the change in predicted product placement after the tariff into an attribute-driven component and a ranking-driven component.

6 Structural

Model

The reduced-form evidence shows that trade shocks on digital platforms operate through multiple channels beyond prices alone. In this section, we develop a simple structural framework to quantify the demand and welfare implications of these adjustments. The model adds the fourth transmission channel in the paper—changes in consumer demand—and allows us to evaluate how demand shifts, product turnover, changes in product attributes, and platform allocation jointly shape consumer and seller outcomes.

The timing is as follows. First, the seller sets the product’s price. Second, the platform observes prices and product characteristics and determines search placement. Finally, consumers observe the products displayed for a given keyword on a given day, together with their prices, positions, and other characteristics, and decide whether and which product to purchase.

6.1 Consumer

We model demand using a simple logit specification in which a market m is a keyword–day pair.

Within each market, consumers choose among the products shown on the first page of search results and an outside option corresponding to not purchasing any product from that page. Let j index products and let $m = (k, t)$ denote the keyword–day market. Consumer i ’s indirect utility from product j in market m is

$$u_{ijm} = \begin{cases} -\alpha_0 p_{jm} + \beta_0 r_{jm} + \mathbf{X}'_{jm} \gamma_0 + \mu_j + \lambda_k + \xi_{jm} + \varepsilon_{ijm}, & \text{Post} = 0, \\ -\alpha_1 p_{jm} + \beta_1 r_{jm} + \mathbf{X}'_{jm} \gamma_1 + \mu_j + \lambda_k + \xi_{jm} + \varepsilon_{ijm}, & \text{Post} = 1, \end{cases} \quad (6.1)$$

where p_{jm} denotes price, r_{jm} denotes search position, and $\mathbf{X}_{jm}$ is a vector of observed product characteristics. The terms μ_j and λ_k denote ASIN and keyword fixed effects, respectively, ξ_{jm} captures time-varying unobserved product–market quality, and ε_{ijm} follows a Type I extreme value distribution. We allow the demand parameters to differ between the pre- and post-tariff periods in order to capture changes in consumer demand after the trade shock.¹⁰

Under the logit assumption, the share of product j relative to the outside option in regime $\tau \in \{0, 1\}$ satisfies

$$\ln s_{jm} - \ln s_{0m} = -\alpha_\tau p_{jm} + \beta_\tau r_{jm} + \mathbf{X}'_{jm} \gamma_\tau + \mu_j + \lambda_k + \xi_{jm}. \quad (6.2)$$

¹⁰Tariffs may change price sensitivity or the valuation of observable product characteristics such as badges, ratings, review counts, and position. Allowing tariff-regime-specific coefficients avoids imposing a common demand system across periods.

Implementing this model in the data requires three additional steps. First, because the data report sales ranks rather than quantities, we convert sales ranks into sales using the mapping procedure in [He and Hollenbeck \(2020\)](#). Second, because sales ranks are observed at the platform rather than the keyword level, we allocate product-level sales across keywords using relative search volume. For each product, we compute the share of total search volume attributable to the focal keyword relative to the total search volume across all keywords in which the product appears, and we use that share to apportion platform-level sales to the keyword level.¹¹ Third, following [Yu \(2024\)](#), we calibrate market size using industry-level benchmarks. Together, these steps yield keyword–day product-level market shares. Appendix [A.13](#) provides additional details.

Price may be endogenous if it is correlated with unobserved product–market quality, ξ_{jm} . To identify α_τ , we instrument for price using interactions among fluctuations in the U.S.–China exchange rate, the daily Trade Policy Uncertainty Index, and the number of China-origin products in the keyword market. Because a substantial share of sellers are based in China, these interactions plausibly shift marginal costs through exchange-rate and policy exposure. Conditional on product and market fixed effects, they are less likely to affect U.S. consumer demand except through prices. This generates exogenous within-market variation in price.

6.1.1 Results

We estimate Equation (6.2) by instrumental variables with keyword and product fixed effects and standard errors clustered at the product level.

Table [A.10](#) reports the demand estimates, and Figure 8 summarizes several implications of the model. Consistent with price endogeneity, the OLS coefficients understate the magnitude of price sensitivity. The IV estimates imply economically meaningful demand elasticities, with most product-level own-price elasticities lying between 3 and 5 in absolute value. These magnitudes are broadly consistent with existing estimates for online retail markets ([Gutierrez, 2021](#); [Farronato et al., 2023](#); [Yu, 2024](#)). We also find that demand becomes less price elastic in the post-tariff period, suggesting that consumers become less responsive to price changes after the shock. One interpretation is that tariff uncertainty changed consumers’ intertemporal purchasing incentives, leading some consumers to purchase immediately rather than delay when future prices were uncertain. This mechanism is consistent with contemporaneous remarks by Amazon’s CEO, who noted that “people have not stopped buying” and that, in some categories, Amazon observed consumers “buying ahead” ([Palmer, 2025](#)).

Search position remains an important determinant of demand. To examine how product utility varies across positions, we compute average normalized mean utility by rank separately in the pre-

¹¹Search-volume data are available for a subsample of 25 adjacent keywords, beginning from a randomly selected starting keyword. The structural analysis is therefore conducted on that subsample.

and post-tariff periods. In both periods, products shown in better positions have higher normalized utility on average, but this gradient becomes flatter after the tariff shock. From the consumer's perspective, the top of the page therefore becomes a weaker signal of product quality.

A similar pattern appears when we examine heterogeneity by badges and consumer feedback. Products carrying the “Best Seller” or “Overall Pick” badges have higher normalized utility than comparable non-badged products, but the relationship between utility and position again becomes flatter in the post-tariff period. Differences by star rating are comparatively modest, whereas products with more reviews exhibit systematically higher utility, even in lower positions. This relationship also becomes flatter after the tariff shock.

Taken together, these results indicate that the demand side of the market also shifts after the tariff shock. Combined with the reduced-form evidence from Section 5.1.1 to Section 5.3, they suggest that the first page of search results becomes less sharply sorted by consumer value after the shock, weakening the informational content of top positions.

6.2 Seller

We model sellers as single-product firms that set prices to maximize static profits, taking the platform's ranking rule as given. Seller j 's profits in market m are

$$\pi_{jm}(p_{jm}) = (p_{jm} - mc_{jm}) M_m s_{jm}, \quad (6.3)$$

where M_m denotes market size, s_{jm} is the demand share implied by Equation (6.2), and mc_{jm} is marginal cost.

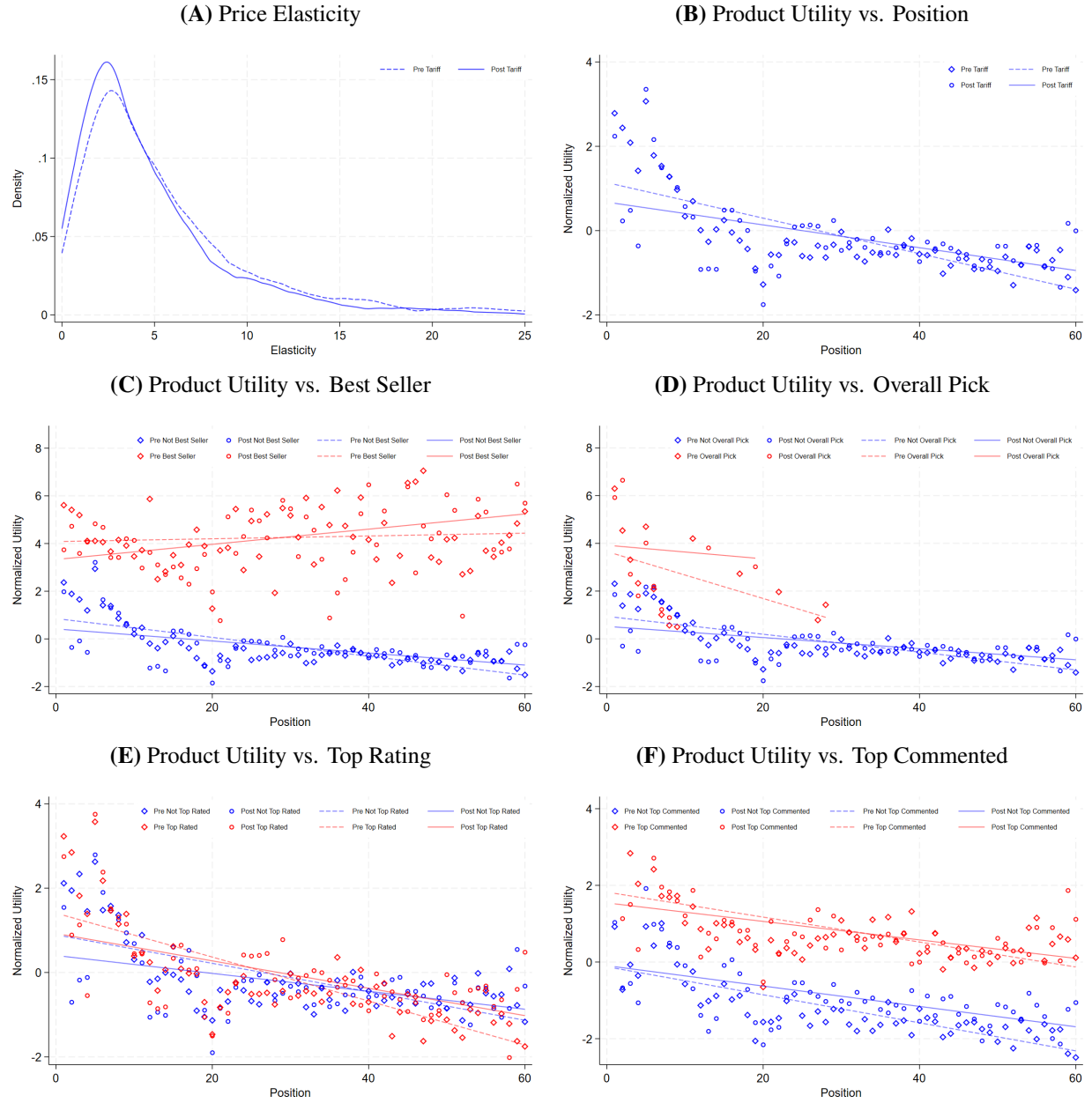
An important feature of the model is that prices affect demand through two channels. They enter utility directly, and they also affect visibility indirectly through the platform's allocation rule. In regime τ , search position is determined by the reduced-form ranking relationship $r_{jm} = f_\tau(p_{jm}, \mathbf{X}_{jm})$. Let α_τ denote the marginal utility of price and β_τ the coefficient on position in mean utility. Differentiating mean utility with respect to price gives

$$\frac{d\delta_{jm}}{dp_{jm}} = -\alpha_\tau + \beta_\tau \frac{\partial f_\tau(\cdot)}{\partial p_{jm}}. \quad (6.4)$$

Under logit demand, product j 's own-price elasticity in market m is

$$\varepsilon_{jjm} = \frac{p_{jm}}{s_{jm}} \frac{ds_{jm}}{dp_{jm}} = p_{jm}(1 - s_{jm}) \left(-\alpha_\tau + \beta_\tau \frac{\partial f_\tau(\cdot)}{\partial p_{jm}} \right). \quad (6.5)$$

Figure 8. Consumer Estimation Results



Notes: This figure summarizes the main demand results. Panel A reports the distribution of own-price elasticities across products and markets. Panel B reports average normalized product utility by position. Panels C–F compare average normalized product utility by position for products with and without the “Best Seller” badge, the “Overall Pick” badge, high ratings, and many reviews.

The seller's first-order condition can therefore be written as

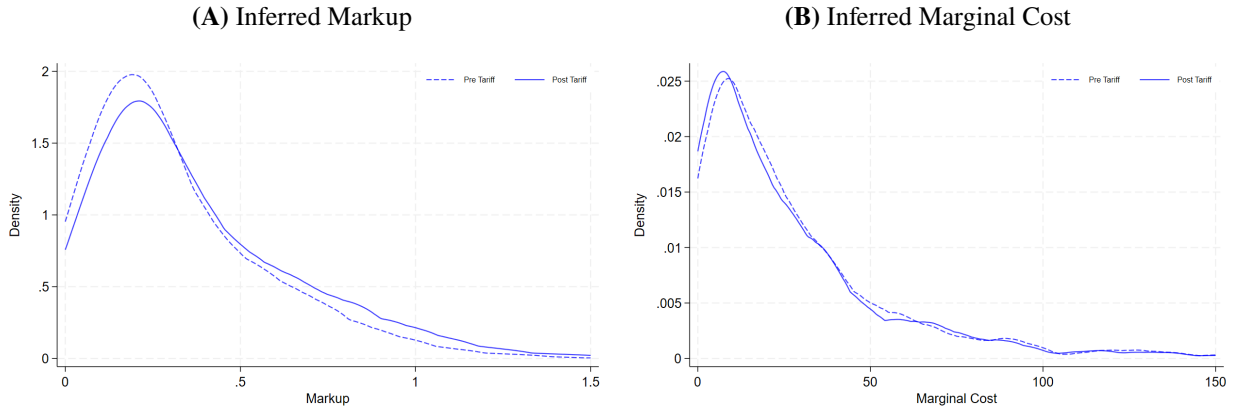
$$\frac{p_{jm} - mc_{jm}}{p_{jm}} = \frac{1}{|\varepsilon_{jgm}|}. \quad (6.6)$$

Combining observed prices with estimated elasticities, we recover implied markups and marginal costs.

6.2.1 Results

Figure 9 plots the distributions of inferred markups and marginal costs before and after the tariff increase. Panel A shows a modest rightward shift in the markup distribution in the post-tariff period, consistent with some increase in pricing margins after the shock. Panel B, by contrast, shows little visible change in the cross-sectional distribution of inferred marginal costs. These distributions should be interpreted conditionally on appearing in the observed sample of first-page search results. Products that exit or lose first-page visibility after the tariff are not observed in the post-tariff period. The post-tariff densities therefore reflect the equilibrium outcomes of surviving visible products rather than the full set of products present before the shock. This selection issue can mask underlying increases in seller costs.

Figure 9. Seller Estimation Results



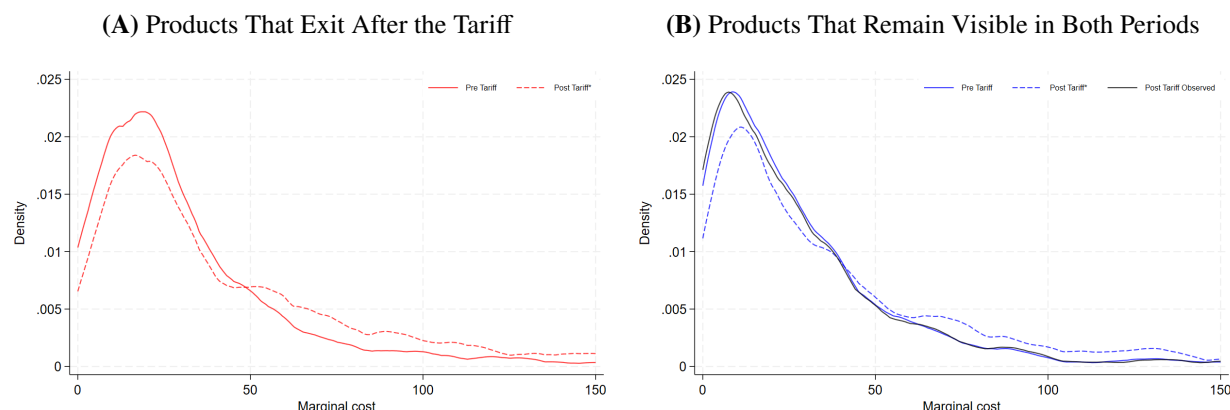
Notes: This figure summarizes the seller-side estimates. Panel A reports the distribution of inferred markups before and after the tariff increase. Panel B reports the corresponding distribution of inferred marginal costs.

Products that disappear from the post-tariff sample are, on average, more expensive in the pre-tariff period than products that remain visible in both periods. If higher pre-tariff prices are correlated with higher marginal costs, the disappearance of these products truncates the upper tail of the observed post-tariff marginal-cost distribution. As a result, even if tariffs raise marginal costs at the product level, the observed cross-section may show limited movement because cost increases among surviving products are offset by the exit or demotion of relatively expensive

products. This selection problem also complicates a simple pass-through comparison, since the composition of observed products changes endogenously after the shock.

To quantify this margin, we supplement the observed distributions with a counterfactual exercise. For products that disappear after the tariff, post-tariff marginal cost is unobserved. We therefore infer pre-tariff marginal cost from the pricing first-order condition and apply the average tariff increase of 112% to construct a counterfactual post-tariff marginal cost under full incidence.

Figure 10. Marginal Costs Before and After the Tariff



Notes: Panel A shows inferred pre-tariff marginal costs for products that disappear after the tariff and the corresponding counterfactual post-tariff marginal cost under full incidence. Panel B shows, for products that remain visible in both periods, inferred pre-tariff marginal cost, observed post-tariff inferred marginal cost, and the counterfactual post-tariff marginal cost under full incidence.

Figure 10 highlights two margins of adjustment. For products that disappear after the tariff, the counterfactual post-tariff marginal-cost distribution shifts sharply to the right, indicating that these products would have faced substantial cost increases under full incidence. For products that remain visible, the observed post-tariff marginal-cost distribution remains much closer to the pre-tariff distribution than to the full-incidence counterfactual. The post-tariff adjustment therefore appears to combine selection, through the disappearance of relatively expensive products, with incomplete transmission of the tariff shock among surviving products.

6.3 Consumer Surplus and Heterogeneous Effects

We now combine the estimated demand system with the observed changes in product sets, product attributes, and platform allocation to quantify the welfare consequences of the tariff shock. The decomposition follows the paper's four-channel structure: demand shifts, product turnover, changes in product attributes, and changes in platform allocation. Because entry and exit are economically distinct, we implement the turnover channel sequentially through separate entry and exit terms.

Let P_τ denote the estimated demand parameters in regime $\tau \in \{0, 1\}$, let J_τ denote the set of available products, let X_τ denote product attributes, and let A_τ denote the platform's ranking environment. Consumer surplus and seller profits are evaluated as functions of these objects.

Consumer surplus in market m is

$$CS_m(J, X, A, P) = \frac{1}{-\alpha(P)} \ln \left(1 + \sum_{j \in J_m} \exp(\delta_{jm}(X, A; P)) \right), \quad (6.7)$$

where $\alpha(P)$ denotes the price coefficient under demand parameter vector P , and $\delta_{jm}(X, A; P)$ denotes product j 's mean utility under product attributes X , platform allocation A , and demand parameters P .

We decompose the change in consumer surplus as

$$\begin{aligned} \Delta CS_m = & \underbrace{\left[CS_m(J_1, X_1, A_1, P_1) - CS_m(J_1, X_1, A_1, P_0) \right]}_{\text{Demand-shift effect}} + \underbrace{\left[CS_m(J_1, X_1, A_1, P_0) - CS_m(J_S, X_1, A_1, P_0) \right]}_{\text{Entry effect}} \\ & + \underbrace{\left[CS_m(J_S, X_1, A_1, P_0) - CS_m(J_S, X_0, A_1, P_0) \right]}_{\text{Attribute effect}} + \underbrace{\left[CS_m(J_S, X_0, A_1, P_0) - CS_m(J_S, X_0, A_0, P_0) \right]}_{\text{Allocation effect}} \\ & + \underbrace{\left[CS_m(J_S, X_0, A_0, P_0) - CS_m(J_0, X_0, A_0, P_0) \right]}_{\text{Exit effect}}. \end{aligned} \quad (6.8)$$

Here J_S denotes the set of surviving products. The entry and exit terms together capture the product-turnover channel. The attribute effect restores pre-tariff attributes for surviving products while holding the post-tariff ranking environment fixed. In the current implementation, X includes prices, badges, ratings, and review counts. The allocation effect then restores the pre-tariff ranking environment while holding surviving products' attributes at their pre-tariff values.

On the seller side, variable profits are

$$\pi_{jm}(J, X, A, P) = (p_{jm}(X) - mc_{jm}(X)) q_{jm}(J, X, A, P), \quad (6.9)$$

where $p_{jm}(X)$ is price, $mc_{jm}(X)$ is inferred marginal cost, and $q_{jm}(J, X, A, P)$ is predicted quantity. Aggregating across products and dividing by market size yields seller profits per consumer,

$$\Pi_m^{pc}(J, X, A, P) = \frac{\sum_{j \in J_m} (p_{jm}(X) - mc_{jm}(X)) q_{jm}(J, X, A, P)}{M_m}, \quad (6.10)$$

where M_m denotes market size.

We evaluate seller profits under the same sequence of counterfactuals. In observed environments,

inferred marginal costs are held at their observed values. In counterfactuals that restore pre-tariff attributes for surviving products, prices and inferred marginal costs are reset to their pre-tariff values. The seller-side exercise is therefore best interpreted as a partial-equilibrium decomposition of how the tariff shock affects sellers through demand shifts, turnover, attribute changes, and platform allocation.

6.3.1 Results

Table 3 reports average consumer surplus per consumer before and after the tariff shock, together with the sequential decomposition of the total change.¹²

Table 3. Consumer Surplus Decomposition

	Initial Level (\$)	Final Level (\$)	Change Level (\$)	Change Percent (%)
	(1)	(2)	(3)	(4)
$\Delta CS: Total$	0.511	0.527	0.016	3.070%
$\Delta CS: Preference$	0.464	0.527	0.063	13.578%
$\Delta CS: Entry$	0.435	0.464	0.029	6.595%
$\Delta CS: Attribute$	0.641	0.435	-0.206	-32.075%
$\Delta CS: Algorithm$	0.705	0.641	-0.064	-9.035%
$\Delta CS: Exit$	0.511	0.705	0.193	37.778%

Notes: This table reports average consumer surplus per consumer at the keyword-generation level, computed using the log-sum formula implied by the estimated demand system. For the total row, columns (1) and (2) report observed pre- and post-tariff levels. For the decomposition rows, columns (1) and (2) report the two objects compared at each step of the sequential decomposition. The preference effect compares post-tariff outcomes under post- and pre-tariff preferences. The entry effect captures post-tariff entrants. The attribute effect restores pre-tariff product attributes for surviving products. The platform-allocation effect restores the pre-tariff ranking rule conditional on pre-tariff attributes. The exit effect compares the rolled-back post-tariff survivor market to the full pre-tariff market. Percent changes are computed relative to column (1).

Consumer surplus rises modestly, from \$0.511 before the tariff to \$0.527 after the tariff, an increase of \$0.016, or 3.1%. This modest net gain masks substantial offsetting forces. Holding demand parameters fixed at their pre-tariff values, the post-tariff market would generate consumer surplus of only \$0.464, implying a positive demand-shift component of \$0.063. Entry also contributes positively: adding post-tariff entrants raises consumer surplus from \$0.435 in the survivor-only post-tariff market to \$0.464 in the full post-tariff market, a gain of \$0.029. By contrast, restoring pre-tariff product attributes for surviving products raises consumer surplus from \$0.435 to \$0.641, implying that observed post-tariff changes in prices and other product characteristics reduce consumer surplus by \$0.206 relative to this counterfactual. Restoring the

¹²Analogous results for seller profits are reported in Appendix A.15.

pre-tariff allocation environment further raises consumer surplus from \$0.641 to \$0.705, implying an additional loss of \$0.064 due to post-tariff platform allocation. Finally, comparing this rolled-back survivor market to the full pre-tariff market yields an exit component of \$0.193.

Taken together, the decomposition implies that the net change in consumer surplus reflects positive demand-shift and turnover effects that are partly offset by deterioration in the attributes and visibility of continuing products. In other words, the tariff shock improves welfare through cheaper turnover, but worsens it through weaker product characteristics and a less favorable allocation of visibility.

To interpret these welfare results, Table 4 compares product characteristics in the full sample, among surviving products, and across exiting and entering products.

Table 4. Heterogeneous Effects Across Product Groups

	Full Sample			Survivors			Survivors vs. Exiters			Exiters vs. Entrants		
	Pre	Post	Diff.	Pre	Post	Diff.	Survivor	Exiter	Diff.	Exiter	Entrant	Diff.
<i>Price</i>	3.354	3.343	-0.011***	3.298	3.299	0.001	3.380	3.653	0.273***	3.653	3.584	-0.069***
<i>Prime</i>	0.883	0.512	-0.371***	0.901	0.536	-0.365***	0.876	0.783	-0.093***	0.783	0.412	-0.371***
<i>Best Seller</i>	0.038	0.037	-0.000	0.042	0.042	0.000	0.027	0.014	-0.013***	0.014	0.013	-0.000
<i>Overall Pick</i>	0.019	0.018	-0.001**	0.022	0.022	-0.000	0.012	0.001	-0.011***	0.001	0.001	0.000
<i>Rating</i>	4.465	4.461	-0.005***	4.471	4.465	-0.006***	4.458	4.434	-0.023***	4.434	4.433	-0.001
<i>Comments</i>	5.955	5.632	-0.323***	6.159	5.814	-0.345***	5.712	4.777	-0.935***	4.777	4.624	-0.153***
<i>Sales Rank</i>	10.274	10.307	0.032***	10.106	10.144	0.038***	10.498	11.183	0.685***	11.183	11.176	-0.007

Notes: Columns (1)–(3) report pre- and post-tariff means in the full sample. Columns (4)–(6) report the corresponding comparison among products observed in both periods. Columns (7)–(9) compare surviving and exiting products in the pre-tariff period. Columns (10)–(12) compare exiting products in the pre-tariff period with entering products in the post-tariff period. Difference columns report the difference in means. Significance stars are based on two-sided t-tests:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Three patterns stand out. First, the positive turnover terms are closely tied to prices. Average prices fall in the full sample, but are essentially unchanged among continuing products, while entrants are cheaper than exiters. The gain from product turnover therefore operates primarily through reallocation toward cheaper products rather than repricing among incumbents. From the platform’s perspective, this pattern is consistent with visibility allocation acting as an effective mechanism for dampening price pass-through from sellers to consumers.

Second, the turnover gains are partly offset by deterioration in product attributes and platform allocation. Among continuing products, Prime prevalence falls sharply, review counts decline, ratings fall slightly, and sales ranks worsen. At the same time, the post-tariff ranking environment becomes less favorable, making prominent placement harder to obtain for both surviving and entering products.

Third, the comparison between surviving and exiting products in the pre period shows which incumbents are most exposed to displacement. Products that later disappear are more expensive,

less likely to be Prime-eligible or badge-bearing, have weaker ratings and fewer reviews, and occupy worse sales ranks before the shock. This pattern indicates that the shock disproportionately displaces relatively weak incumbent products.

7 Conclusion

This paper studies how trade shocks transmit through digital marketplaces. We focus on the sharp escalation of U.S. tariffs on Chinese imports in April of 2025, which also generated a substantial increase in trade policy uncertainty. Using high-frequency product-level data from Amazon, we show that the effects of this episode cannot be understood through prices alone. In digital markets, trade shocks propagate through four related channels: product access and turnover, prices and non-price product attributes, platform allocation of visibility, and consumer demand.

The reduced-form evidence establishes the first three channels. First, the increase in tariffs and the heightened trade policy uncertainty reduces the early-life visibility of newly introduced Chinese products, consistent with a worsening of the cold-start problem for exposed sellers. Second, the tariff shock generates incomplete retail price pass-through together with substantial adjustment in non-price margins, including Prime eligibility, badges, sponsored placement, and the composition of products shown to consumers. Third, the platform’s ranking environment changes after the tariff shock, so that the mapping from product characteristics to visibility becomes less favorable for affected products. These findings imply that the platform is not a passive conduit for the shock. It actively mediates how higher trade costs translate into consumer exposure and seller outcomes. The structural analysis adds the fourth channel by showing that consumer demand also shifts after the tariff shock. Search position remains strongly associated with demand, but the relationship between position and estimated consumer value becomes flatter in the post-tariff period. From the consumer’s perspective, the top of the page therefore becomes a weaker signal of product quality. On the seller side, we find incomplete pass-through together with strong evidence of selection: products that disappear after the tariff are, on average, more expensive before the shock than products that remain visible. As a result, the observed post-tariff distribution of marginal costs understates the full increase in cost pressure generated by the tariff.

The welfare decomposition brings these results together. Consumer surplus changes only modestly on net, but this masks substantial offsetting forces. Product turnover shifts the displayed product mix toward cheaper products, which raises consumer surplus. At the same time, these gains are partly offset by deterioration in the attributes of continuing products and by a less favorable post-tariff allocation of visibility. The heterogeneous analysis shows that prices fall mainly through turnover rather than repricing among incumbents, while Prime prevalence, review counts, ratings, and sales ranks worsen even within the set of surviving products. Taken together,

the results show that the incidence of the tariff shock depends on how demand shifts, product turnover, product attributes, and platform allocation interact.

More broadly, the paper shows that understanding trade in the digital economy requires moving beyond conventional pass-through frameworks. Platforms influence not only the prices consumers pay, but also which products consumers encounter, which sellers obtain prominent exposure, and how demand is reallocated across the marketplace. Visibility allocation can be an effective instrument to dampen price pass-through, but it also reshapes product composition and the distribution of seller exposure. In platform-mediated markets, the incidence and welfare effects of trade policy therefore depend jointly on seller pricing, product turnover, platform allocation, and consumer demand. As digital platforms account for a growing share of retail activity, trade policy will increasingly affect markets not only through prices, but also through the platform-mediated allocation of consumer attention.

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Appendices

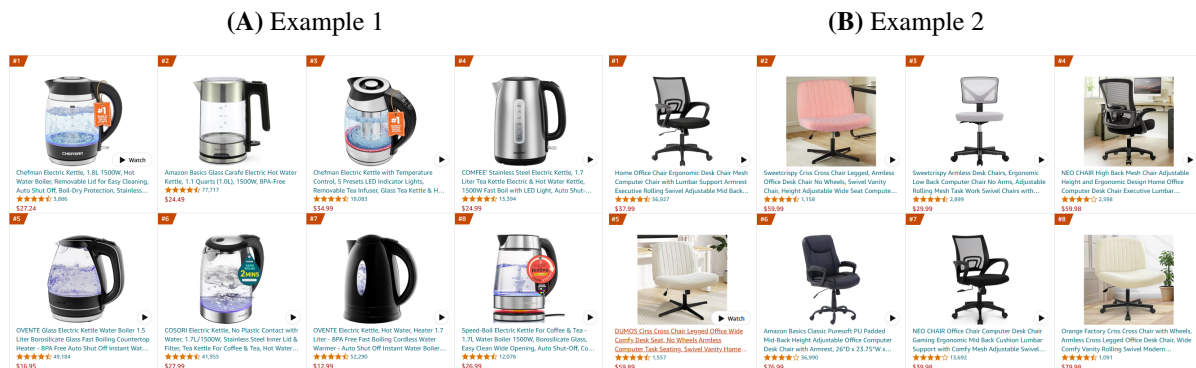
A.1 Product

Information

This section provides additional detail on the construction of the product sample and on the comparability of products within narrow Amazon subcategories. Because the main analysis treats narrow subcategory names as keyword-level product markets, it is important to show that products grouped within a keyword are meaningfully comparable. We therefore present illustrative examples of products within narrowly defined subcategories, describe the text-similarity measure used in the data section, and show examples of the interfaces from which product information is scraped.

Figure A.1 illustrates examples of top-selling products within narrow subcategories of the “Home & Kitchen” category on Amazon.

Figure A.1. Example of Product Similarity within Subcategory



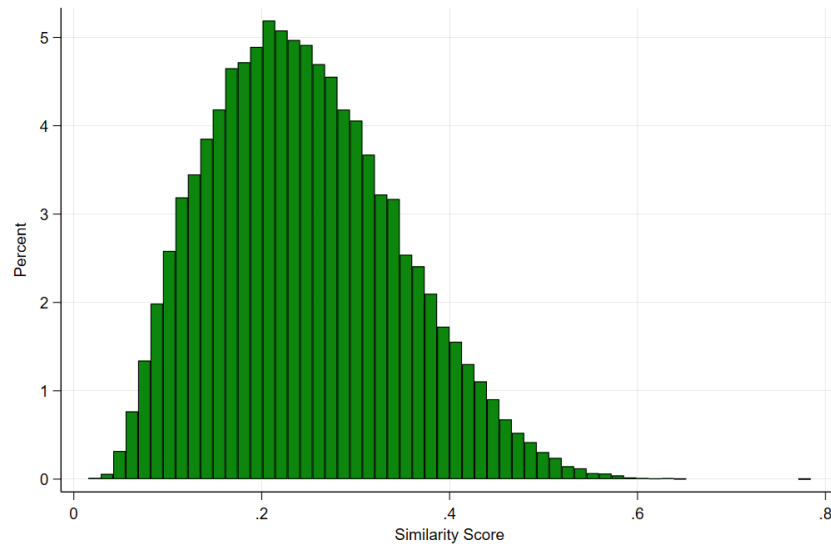
Notes: The figure illustrates examples of top-selling products within narrow subcategories of the “Home & Kitchen” product category on Amazon. Panel A shows products from the “Electric Kettle” subcategory, while Panel B displays products from the “Office Chair” subcategory.

To assess comparability more systematically, we tokenize product titles into individual words and adjacent word pairs and compute a similarity score between 0 and 1 by comparing each product’s tokens with those of other products on the first search-results page for the same keyword. Higher values indicate that a product is more similar to the other products shown under the same keyword. Figure A.2 reports the resulting distribution of similarity scores.

Figure A.3 provides an example of products with low and high similarity scores for a specific keyword. The item highlighted in the green box has a similarity score of about 0.7, while the item highlighted in the red box has a similarity score of about 0.1. This example clarifies how the similarity measure distinguishes closely related products from more weakly related listings.

Finally, Figure A.4 provides examples of the interfaces from which we scrape product

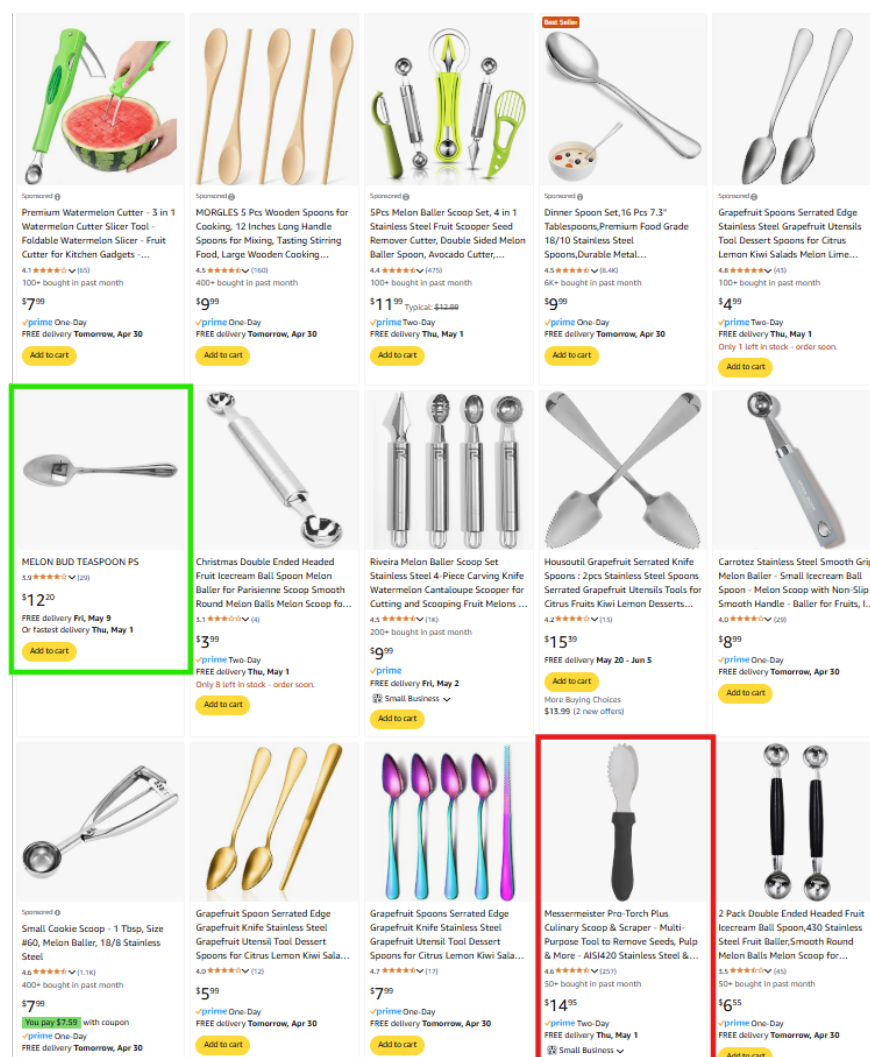
Figure A.2. Similarity Score Distribution



Notes: This figure illustrates the distribution of similarity scores for products appearing on the first search-results page.

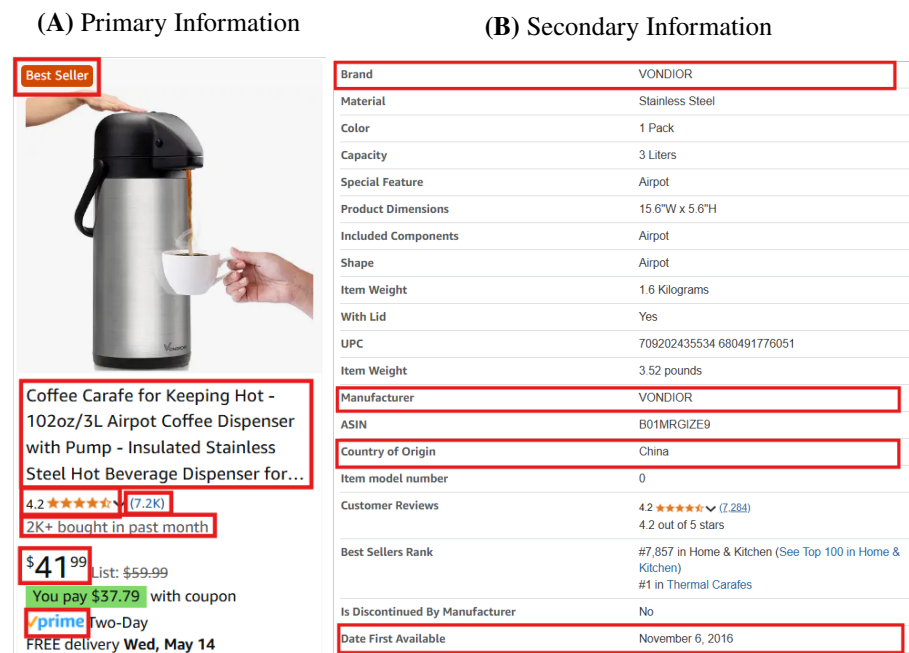
information. These screenshots illustrate the distinction between the primary product attributes used in the main analysis, such as title and price, and secondary information, such as seller details and fulfillment-related information.

Figure A.3. Similarity Score Example



Notes: This figure illustrates an example of a product with a low similarity score (red box) and a high similarity score (green box) on the first results page for the keyword “Melon Spoons.”

Figure A.4. Example of Product Information



Notes: The figure provides examples of product information extracted from online listings, distinguishing between primary attributes (e.g., title) and secondary attributes (e.g., seller details).

A.2 Summary

Statistics

This section reports summary statistics for the scraped product-level dataset used in the main analysis. The table summarizes the main price, sales, fulfillment, and review variables and provides a compact overview of the cross-sectional variation in the sample.

Table A.1. Summary Statistics

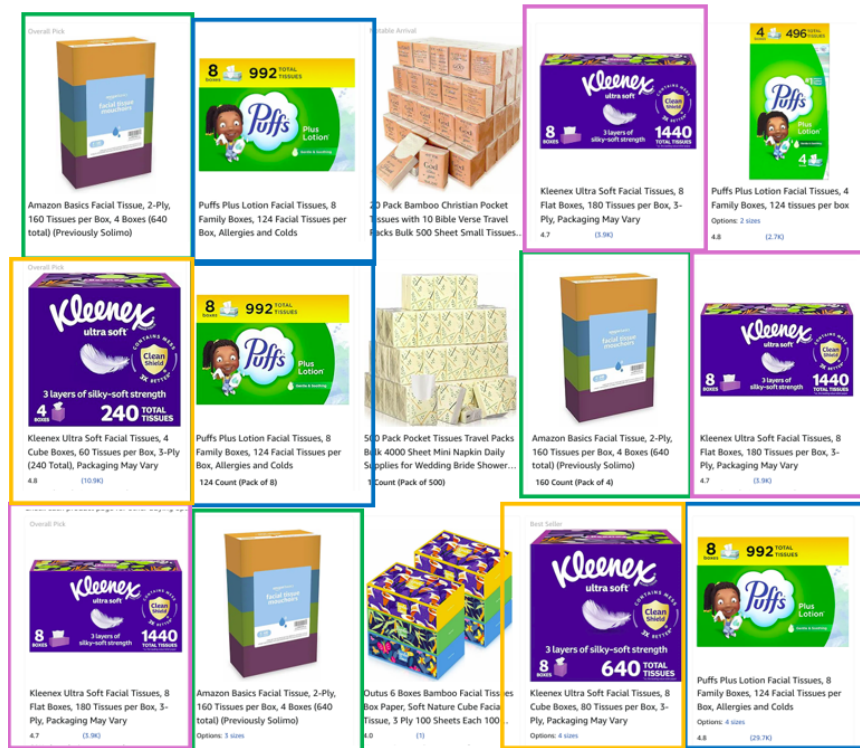
	Obs. (1)	Min. (2)	Mean (3)	Median (4)	Max. (5)	S.D. (6)
Sales Rank	1,195,986	1.00	160,996.00	35,183.00	13,118,728.00	431,543.80
Sales Last Month	880,732	50.00	1,362.93	400.00	100,000.00	3,918.82
Price (US \$)	1,209,818	0.00	49.81	23.99	999.99	85.64
Referral Fee (%)	1,180,495	4.93	14.81	15.01	45.03	1.14
Star Rating (1-5)	1,168,749	1.00	4.46	4.50	5.00	0.33
Comments	1,080,430	0.00	3,370.63	373.00	849,510.00	13,529.69
Prime	1,226,532	0.00	0.73	1.00	1.00	0.44

Notes: This table reports summary statistics. A lower sales rank corresponds to higher sales. Sales last month is a categorical variable indicating the product's sales volume in the previous month. Price is measured in U.S. dollars. Referral fee is expressed as a percentage. Star Rating is an ordinal measure ranging from 1 (lowest) to 5 (highest). Comments refers to the number of user-generated product reviews. Prime is a binary indicator equal to 1 if the product is Amazon Prime and 0 otherwise

A.3 Personalization

Amazon search rankings may vary across users or browsing sessions. This section provides a simple illustration of that variation. The purpose is not to model personalization directly, but rather to document that the same keyword can generate somewhat different rankings across sessions, which is relevant when interpreting placement as an equilibrium outcome of the platform’s ranking environment.

Figure A.5. Personalization Example



Notes: This figure illustrates variation in product rankings for the same search keyword across different user sessions. Highlighted boxes indicate identical products appearing in different positions.

A.4 Product Distribution by Country

This section reports the full distribution of observed countries of origin in the sample. It complements the summary figure in the main text by showing the complete country-level breakdown, both in counts and in shares, for all products for which country-of-origin information is available.

Table A.2. Distribution by Country of Origin

Country	No.	%	Country	No.	%
Afghanistan	95	0.022%	Korea, Republic of	680	0.159%
Albania	47	0.011%	Kyrgyzstan	19	0.004%
Algeria	8	0.002%	Lao People's D.R.	22	0.005%
American Samoa	24	0.006%	Latvia	15	0.004%
Andorra	29	0.007%	Lithuania	5	0.001%
Anguilla	72	0.017%	Macau	14	0.003%
Antarctica	7	0.002%	Malaysia	677	0.158%
Argentina	36	0.008%	Mexico	1380	0.322%
Armenia	55	0.013%	Morocco	225	0.053%
Australia	34	0.008%	Namibia	1	0.000%
Austria	106	0.025%	Netherlands	154	0.036%
Bangladesh	56	0.013%	New Zealand	27	0.006%
Belgium	67	0.016%	Niue	9	0.002%
Bouvet Island	2	0.000%	Pakistan	3359	0.784%
Brazil	213	0.050%	Philippines	18	0.004%
Brunei Darussalam	2	0.000%	Poland	442	0.103%
Bulgaria	42	0.010%	Portugal	175	0.041%
Cambodia	61	0.014%	Puerto Rico	20	0.005%
Canada	453	0.106%	Romania	355	0.083%
C.A.R.	25	0.006%	Russian Federation	13	0.003%
Chile	147	0.034%	Singapore	139	0.032%
China	370349	86.438%	Slovakia	172	0.040%
Christmas Island	4	0.001%	Slovenia	83	0.019%
Colombia	256	0.060%	South Africa	1	0.000%
Croatia	6	0.001%	Spain	424	0.099%
Czech Republic	74	0.017%	Sri Lanka	34	0.008%
Denmark	76	0.018%	St. Helena	4	0.001%
Dominican Republic	1	0.000%	Sweden	107	0.025%
Ecuador	4	0.001%	Switzerland	624	0.146%
Egypt	164	0.038%	Taiwan	3054	0.713%
El Salvador	47	0.011%	Thailand	1595	0.372%
Estonia	17	0.004%	Tunisia	165	0.039%
F.S. of Micronesia	13	0.003%	Turkey	1624	0.379%
Finland	17	0.004%	USA	22617	5.279%
France	691	0.161%	Ukraine	522	0.122%
Georgia	18	0.004%	United Arab Emirates	38	0.009%
Germany	1129	0.264%	United Kingdom	160	0.037%
Ghana	5	0.001%	Venezuela	24	0.006%
Greece	7	0.002%	Vietnam	3654	0.853%
Hong Kong	234	0.055%	Hungary	106	0.025%
India	7286	1.701%	Ireland	4	0.001%
Indonesia	617	0.144%	Israel	373	0.087%
Italy	1273	0.297%	Jamaica	19	0.004%
Japan	1432	0.334%			

Notes: This table reports the distribution of products by country of origin among products for which country-of-origin information is either displayed or recovered.

A.5 Timeline of Major U.S. Tariff Measures in 2025

This section summarizes the major 2025 trade-policy actions between the United States and China that are relevant for the empirical analysis. The table is intended to clarify the timing of the policy shocks used in the paper and to distinguish the April tariff escalation from the later *de minimis* changes and the subsequent partial easing.

Table A.3. Major Trade Policy Events Between the U.S. and China in 2025

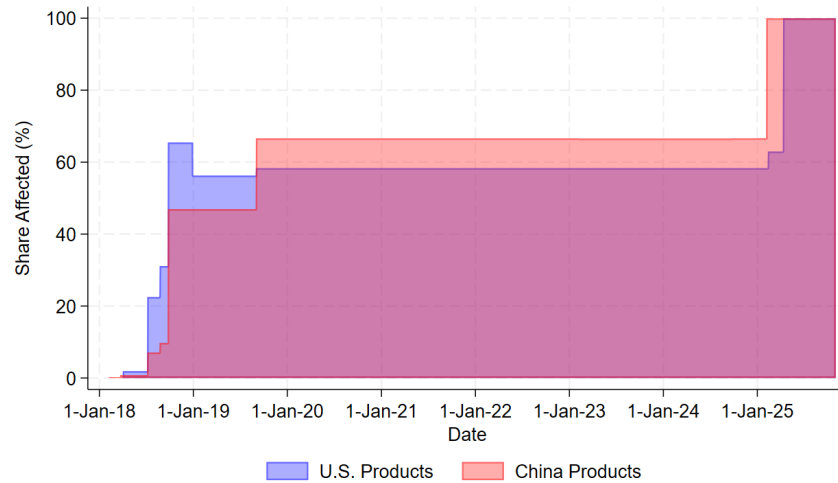
Date	Country	Event
<i>February 4, 2025</i>	US	10% tariff imposed on all imports from China.
<i>February 10, 2025</i>	China	China imposes retaliatory tariffs.
<i>March 4, 2025</i>	US	Tariffs on imports from China increased from 10% to 20%.
<i>March 10, 2025</i>	China	China responds with additional tariffs.
<i>April 10, 2025</i>	US	Tariffs on imports from China raised to 125%.
<i>April 10, 2025</i>	China	China enacts corresponding tariff increases.
<i>May 14, 2025</i>	US	Tariffs on imports from China reduced from 125% to 10%.
<i>May 14, 2025</i>	China	Tariffs on U.S. imports reduced from 125% to 10%.

Notes: This table summarizes key 2025 trade policy actions between the U.S. and China. Source: [Trump's Trade War Timeline 2.0: An Up-to-Date Guide](#).

A.6 Share of Affected Products

This section provides additional context on the breadth of tariff exposure. Figure A.6 plots the share of Chinese exports covered by U.S. tariffs and the share of U.S. exports covered by Chinese tariffs over time. By early 2025, nearly all Chinese exports to the United States were covered by tariffs, indicating that the policy shock was both broad and economically meaningful.

Figure A.6. Affected Share Trends Timeline

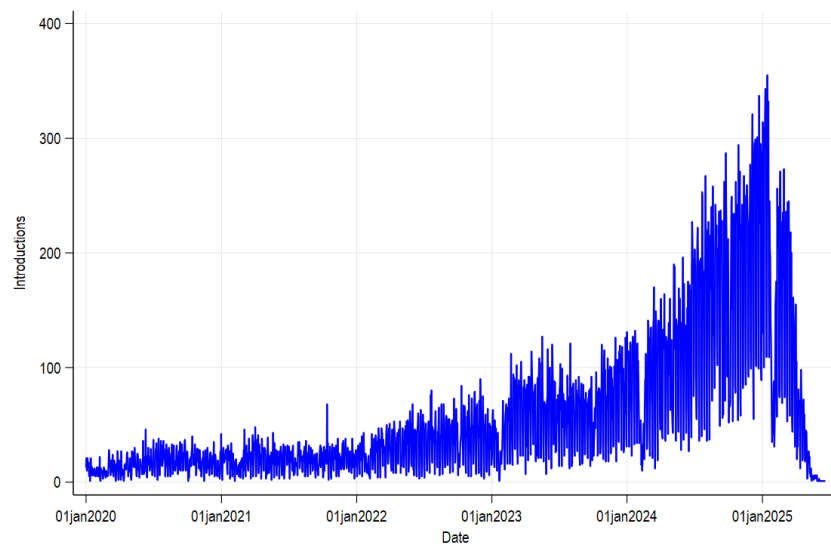


Notes: This figure plots the share of Chinese products affected by U.S. tariffs and the share of U.S. products affected by Chinese tariffs over time. Data are from [Bown \(2025\)](#).

A.7 Available Since Time Series

This section provides additional descriptive evidence on the timing of product introductions. Figure A.7 plots the number of products on the first search-results page by listing date. The figure complements the event-study evidence in the main text by illustrating time variation in the arrival of newly introduced products.

Figure A.7. Available Since Time Series



Notes: This figure shows the number of products available for each date in the sample period. The x-axis reports the date, and the y-axis reports the number of available products.

A.8 Robustness Check: Pre-Tariff Shock Regression

This section assesses the plausibility of the parallel-trends assumption underlying the main difference-in-differences specification. We do so by estimating pre-tariff regressions that interact the high-exposure indicator with generation dummies while including keyword fixed effects. The goal is to test whether more- and less-exposed product markets exhibit systematically different trends before the tariff shock.

To assess the validity of the parallel-trends assumption underlying our DiD, we estimate a pre-tariff shock regression of variables of interest on the interaction between the High-Exposure indicator and the time dummies, including keyword fixed effects:

$$y_{ct} = \alpha + \beta \text{High-Exposure}_{ct} + \gamma \text{Generation}_t + \delta (\text{High-Exposure}_{ct} \times \text{Generation}_t) + \kappa_c + \varepsilon_{ct}. \quad (\text{A.1})$$

This specification isolates differences between High-Exposure and Low-Exposure product categories within each generation prior to the tariff shock. The pre-tariff regression results in Table A.4 show little evidence of systematic pre-shock divergence in prices or in most other product characteristics. This supports the interpretation of the main difference-in-differences estimates as capturing post-shock changes rather than pre-existing trends.

Table A.4. Pre-Tariff Shock Regression Results

	Price	Prime Delivery	Best Seller	Overall Pick	Sales Rank	Ratings	Com- ments	Brand Variety	Manuf. Variety	# Sponsored
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>High-Exposure</i> $\times t = 2$	-0.011 (0.012)	0.001 (0.003)	0.002 (0.002)	0.003** (0.001)	-0.020 (0.036)	0.001 (0.001)	0.025 (0.025)	-0.029 (0.019)	-0.032 (0.019)	-0.014 (0.015)
<i>High-Exposure</i> $\times t = 3$	0.007 (0.012)	0.004 (0.004)	0.001 (0.002)	0.003*** (0.001)	0.006 (0.037)	0.001 (0.001)	0.002 (0.030)	-0.047*** (0.016)	-0.048*** (0.016)	-0.018 (0.012)
<i>High-Exposure</i> $\times t = 4$	-0.016 (0.013)	-0.001 (0.004)	0.002 (0.002)	0.003** (0.001)	0.053 (0.038)	0.000 (0.001)	0.024 (0.029)	-0.051*** (0.016)	-0.049*** (0.016)	-0.023** (0.012)
<i>High-Exposure</i> $\times t = 5$	-0.023* (0.012)	0.005 (0.004)	0.001 (0.002)	0.002 (0.001)	0.039 (0.037)	0.000 (0.001)	0.012 (0.032)	-0.038** (0.016)	-0.040** (0.016)	-0.013 (0.013)
<i>High-Exposure</i> $\times t = 6$	-0.011 (0.013)	0.000 (0.004)	0.001 (0.002)	0.003** (0.001)	-0.038 (0.039)	-0.001 (0.001)	0.017 (0.032)	-0.060*** (0.016)	-0.056*** (0.016)	-0.026** (0.012)
<i>High-Exposure</i> $\times t = 7$	-0.006 (0.014)	-0.001 (0.004)	-0.000 (0.002)	0.003** (0.001)	0.031 (0.044)	-0.000 (0.001)	0.010 (0.030)	-0.039** (0.016)	-0.041** (0.016)	-0.029** (0.012)
<i>High-Exposure</i> $\times t = 8$	-0.006 (0.014)	0.000 (0.004)	0.001 (0.002)	0.002 (0.001)	-0.020 (0.046)	0.000 (0.001)	0.042 (0.032)	-0.043** (0.017)	-0.048*** (0.017)	-0.029** (0.012)
<i>High-Exposure</i> $\times t = 9$	0.003 (0.014)	-0.002 (0.005)	-0.000 (0.002)	0.004*** (0.001)	-0.021 (0.042)	-0.000 (0.001)	-0.008 (0.031)	-0.047*** (0.017)	-0.046*** (0.017)	-0.068*** (0.015)
<i>High-Exposure</i> $\times t = 10$	-0.008 (0.013)	-0.007 (0.004)	0.002 (0.002)	0.002 (0.001)	0.033 (0.043)	0.000 (0.001)	0.033 (0.037)	-0.040** (0.017)	-0.039** (0.017)	-0.032*** (0.012)
Keyword FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,820	12,820	12,820	12,820	12,820	12,820	12,817	12,820	12,820	12,692

Notes: This table shows the estimates from Equation A.1. γ estimates are excluded in the table. Standard errors are clustered at the keyword-level and are in parenthesis. * p<0.1, ** p<0.05, *** p<0.01.

A.9 Robustness Check: Alternative Exposure Cutoff

This section evaluates whether the main reduced-form results are sensitive to the definition of exposure. In the main text, a category is classified as high exposure if it contains more than 14 China-origin products, corresponding to the 25th percentile of the exposure distribution. Here we use a stricter definition based on 24 China-origin products, corresponding to the 50th percentile.

As a robustness check, we redefine a category as high exposure using a higher cutoff: 24 China-origin products rather than 14, corresponding to the 50th percentile of the exposure distribution instead of the 25th percentile. This alternative definition allows us to assess whether the main results are sensitive to the choice of cutoff. The findings remain qualitatively similar, indicating that the results are not driven by this particular threshold choice.

Table A.5. Tariff Results

	Price	Prime Delivery	Best Seller	Overall Pick	Sales Rank	Ratings	Com- ments	Brand Variety	Manuf. Variety	# Sponsored
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>High-Exposure</i>	-0.019*** (0.006)	0.049*** (0.010)	-0.003** (0.001)	-0.002*** (0.000)	0.039* (0.020)	-0.001 (0.000)	-0.024 (0.024)	0.105*** (0.014)	0.102*** (0.014)	0.080*** (0.012)
<i>Post</i>	-0.026*** (0.005)	-0.364*** (0.004)	0.000 (0.001)	-0.001** (0.000)	-0.008 (0.011)	-0.001*** (0.000)	-0.390*** (0.015)	0.001 (0.006)	-0.003 (0.006)	-0.006 (0.004)
<i>High-Exposure × Post</i>	0.017*** (0.005)	-0.020*** (0.006)	-0.001 (0.001)	-0.000 (0.000)	0.033** (0.014)	0.000 (0.000)	0.018 (0.018)	0.019*** (0.006)	0.021*** (0.006)	0.011*** (0.004)
Keyword FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,782	21,782	21,782	21,782	21,782	21,782	21,782	21,782	21,782	21,782

Notes: This table shows the estimates from Equation 5.2. Columns (1) shows results related to price. Columns (2)–(10) show results related to delivery, platform-assigned badges, sales, ratings, comments, brand variety, manufacturer variety, and the number of sponsored products. Robust standard errors are clustered at the keyword level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.10 Robustness Check: Product Level Analysis

This section re-estimates the main reduced-form specification at the product level. The goal is to assess whether the main results are sensitive to the level of aggregation and whether the subcategory-level estimates conceal meaningful within-market heterogeneity. Because country-of-origin information is required at the product level, these regressions are estimated on the subsample of products for which country-of-origin information is observed.

As a robustness check, we replicate the analysis at the product level. This exercise allows us to assess whether the main results are sensitive to the level of aggregation. The findings remain qualitatively similar, indicating that the results are not driven by the keyword-level aggregation. These estimates, however, are conditional on country-of-origin information being observed for the product.

Table A.6. Tariff Transmission Results

	Price (1)	Prime Delivery (2)	Best Seller (3)	Overall Pick (4)	Sales Rank (5)	Rating (1–5) (6)	Com- ments (7)
<i>Post</i>	0.009*** (0.002)	-0.502*** (0.003)	0.004* (0.002)	-0.001 (0.002)	0.063*** (0.003)	-0.001*** (0.000)	0.361*** (0.008)
<i>High-Exposure × Post</i>	0.007*** (0.002)	-0.051*** (0.004)	-0.004* (0.002)	0.000 (0.002)	0.027*** (0.003)	-0.001* (0.000)	-0.002 (0.008)
Keyword FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ASIN FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	609,288	615,302	615,302	615,302	604,447	587,990	549,078

Notes: This table shows the estimates from Equation 5.2 on a product-level. Standard errors are clustered at the ASIN-level and in parenthesis. * p<0.1, ** p<0.05, *** p<0.01.

A.11 Robustness Check: *De Minimis* Termination

This section uses the termination of the *de minimis* exemption as an alternative policy shock. The purpose is to show that the main reduced-form results are not specific to the April 10 tariff increase, but also arise when exposure is defined using a closely related cost shock that directly affected low-value China-origin shipments.

As a robustness check, we exploit the termination of the *de minimis* exemption, which directly affected products valued under \$800 from China and Hong Kong that were previously allowed to enter the U.S. duty-free. With the exemption removed, these products became subject to a tariff of either 120% of their declared value or a flat fee of \$100 per item, whichever is higher. After May 14, 2025, these rates were reduced to 54% and \$100, respectively. This sudden and substantial change serves as an exogenous tariff shock to identify the impact of trade policy.

We re-estimate Equation 5.2 using May 2, 2025—the date the *de minimis* exemption was revoked—as the timing of the exogenous shock. The estimates remain qualitatively similar to the main results, indicating that the main conclusions are not driven by the exact timing of the April 10 tariff increase.

Table A.7. Tariff Results

	Price	Prime Delivery	Best Seller	Overall Pick	Sales Rank	Ratings	Com- ments	Brand Variety	Manuf. Variety	# Sponsored
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>High-Exposure</i>	-0.000 (0.010)	0.054*** (0.011)	-0.007*** (0.002)	-0.004*** (0.001)	0.117*** (0.032)	0.000 (0.001)	-0.062 (0.039)	0.204*** (0.027)	0.200*** (0.027)	0.141*** (0.021)
<i>Post</i>	-0.042*** (0.008)	-0.496*** (0.008)	0.004*** (0.001)	-0.000 (0.001)	-0.041** (0.020)	0.001 (0.001)	0.231*** (0.018)	-0.019 (0.013)	-0.027** (0.013)	-0.020** (0.009)
<i>High-Exposure × Post</i>	0.011 (0.009)	-0.038*** (0.009)	-0.004*** (0.001)	-0.000 (0.001)	0.042* (0.021)	-0.001** (0.001)	-0.001 (0.020)	0.040*** (0.014)	0.045*** (0.013)	0.024*** (0.009)
Keyword FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,782	21,782	21,782	21,782	21,782	21,782	21,778	21,782	21,782	21,607

Notes: This table shows the estimates from Equation 5.2. Columns (1) shows results related to price. Columns (2)–(10) show results related to delivery, platform-assigned badges, sales, ratings, comments, brand variety, manufacturer variety, and the number of sponsored products. Standard errors are clustered at the keyword level and reported in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

A.12

Robustness

Check:

Position

Algorithm

A.12.1

Position

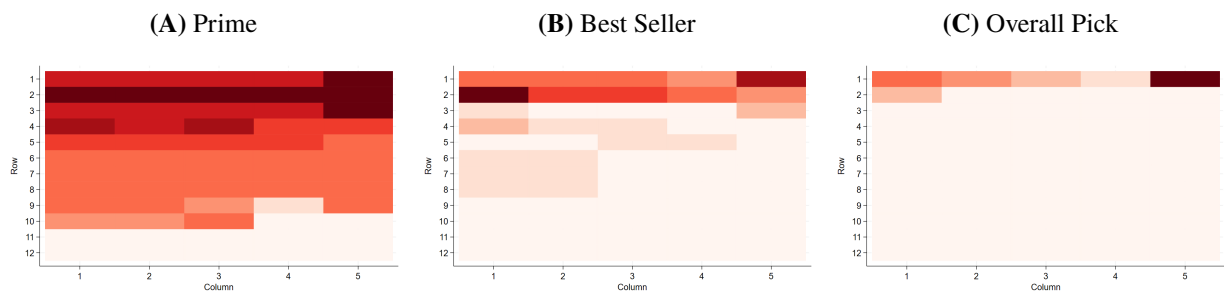
Algorithm

Variables

This section provides descriptive evidence for the variables included in the platform-allocation specification in the main text. The goal is not to claim that these variables exhaust the platform’s ranking rule, but rather to show that badges, ratings, and sales-performance measures are systematically related to observed product placement and are therefore empirically relevant predictors of visibility.

Other than price, our approximation of Amazon’s positioning algorithm incorporates the Best Seller badge, the Overall Pick badge, the Prime badge (our proxy for FBA eligibility), product ratings, and sales rank. Figure A.8 shows how platform-assigned badges are distributed across search positions. If badges contain information used by the platform’s ranking environment, they should appear disproportionately in certain parts of the page. Figure A.9 then shows how ratings and sales rank vary with placement, providing additional descriptive support for including these variables in the reduced-form specification.

Figure A.8. Distribution of Platform Badges Across Search Positions



Notes: This figure shows the distribution of platform-assigned badges across product positions on the search-results page. Each cell reports the frequency with which a given badge appears at a particular position. The concentration of these badges in specific parts of the page suggests that they are systematically associated with product placement.

A.12.2

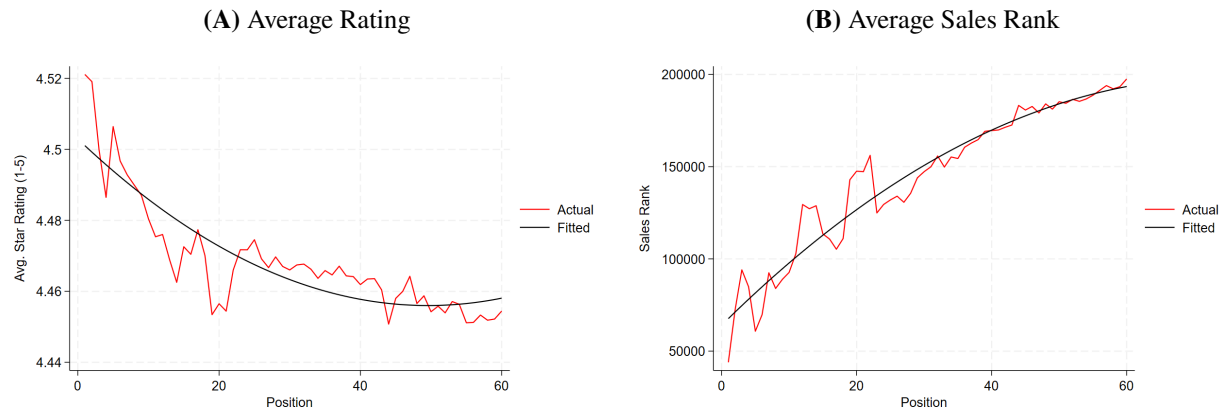
Ranking

Algorithm

Placebo

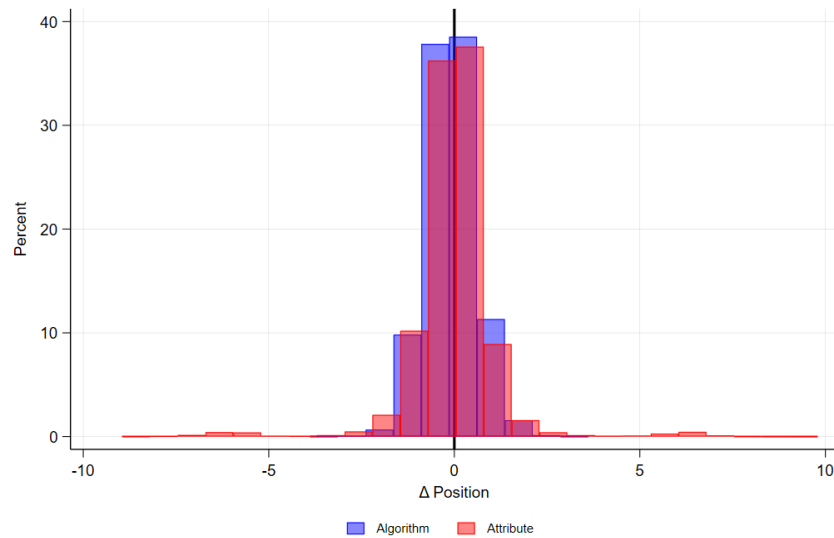
Using an early pre-tariff sample, to ensure that exposure to the tariff was minimal, we examine whether products switched positions as a function of their attributes and rankings. Although the platform’s ranking algorithm may change over time, our focus is on the direction of the resulting reallocation. We find, that compared with the pre-post analysis in the main study, the distribution is more tightly concentrated around zero.

Figure A.9. Product Quality and Performance Measures Across Search Positions



Notes: This figure plots average product ratings and average sales rank by search position. The fitted lines summarize the relationship between these product characteristics and placement. The systematic variation across positions indicates that these variables are informative predictors of product placement and should therefore be included in the approximation of the platform's positioning algorithm.

Figure A.10. Changes in Product Placement



Notes: This figure decomposes the change in predicted product placement using a pre-tariff sample into an attribute-driven component and a ranking-driven component.

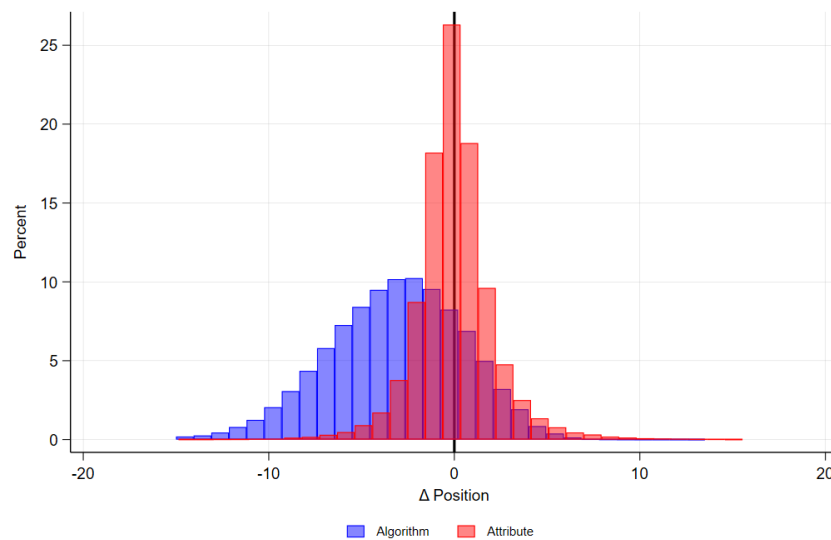
A.12.3 Reverting

Ranking

Algorithm

We also examine whether the ranking algorithm adjusts after the reversal of the tariffs in mid-May 2025. If the tariff increase induced the platform to penalize products more strongly that increased their price, then the subsequent tariff reduction should weaken the incentive to maintain such a restrictive ranking rule. Consistent with this interpretation, we find that after the tariff reduction the algorithm becomes less strict. The distribution of estimated algorithmic changes shifts to the left, indicating that, holding product attributes fixed, the post-reversal ranking rule places products in relatively better positions.

Figure A.11. Changes in Product Placement



Notes: This figure decomposes the change in predicted product placement using a post-tariff sample and using the tariff reduction as a cutoff into an attribute-driven component and a ranking-driven component.

This section provides additional detail on the implementation of the structural model. The main text presents the structural framework in compact form; here we explain more fully how platform-level sales are converted into keyword-level market shares, how market size is calibrated, how the price instrument is used, and how the resulting demand estimates relate to the welfare decomposition. Because the structural results play an important role in the paper’s contribution, the discussion below is intentionally more detailed than the earlier appendix sections.

A.13.1 Calibration of Fraction of Sales From Focal Keywords

The demand model is estimated at the keyword–day level, but the raw sales information is observed at the platform level. This creates a measurement problem because a given product may appear in search results for multiple keywords. If all observed platform-level sales were attributed to the focal keyword, estimated utility would be biased upward for products that sell well elsewhere on the platform but perform weakly in the observed keyword market.

To address this issue, we assign only a fraction of a product’s platform-wide sales to the focal keyword. Let $\kappa_{jt} \in [0, 1]$ denote the share of product j ’s total keyword search volume accounted for by the focal keyword on day t . We assume that this search-volume share maps proportionally into the share of the product’s total sales generated through that keyword. Let q_{jt} denote product j ’s total sales on day t , and let M_t denote the size of the focal keyword market. The implied market share is then

$$s_{jt}(p) = \frac{\kappa_{jt}q_{jt}}{M_t}. \quad (\text{A.1})$$

This adjustment is especially important for products that appear under many related keywords. A product that ranks poorly for the focal keyword may still generate substantial platform-wide sales because it ranks highly under other, more relevant search terms. The κ_{jt} adjustment prevents those outside sales from being incorrectly attributed to the observed market. In this sense, the adjustment is designed to bring the empirical market-share measure closer to the object required by the keyword-level demand model.

A.13.2 Calibration of Market Size

The logit model requires a market size M_t for each keyword–day market. Because the data do not directly report the number of consumers searching a keyword on a given day, we calibrate market size using aggregate information on Amazon sales and search traffic. The goal is to obtain a

plausible estimate of the number of potential buyers corresponding to each keyword–day market rather than to measure total platform traffic directly.

The calibration proceeds in two steps. First, we construct an estimate of total quarterly units sold on Amazon’s marketplace. Amazon’s Q1 and Q2 2025 earnings releases indicate average online-store revenue of \$59.4 billion per quarter, of which about 60% is attributed to North America. Our analysis focuses on Amazon.com, a domain primarily serving the U.S. market. Using the average sales-weighted price in our sample, \$49.81, this implies approximately 0.72 billion first-party units sold in North America per quarter. Since 61.5% of units are sold by third-party sellers, total marketplace unit sales are approximately

$$\frac{0.72}{0.615} \approx 1.2 \text{ billion units per quarter.}$$

Second, we construct an estimate of quarterly keyword searches on Amazon.com. The platform receives roughly 2.3 billion visits in the first quarter of 2024, and we assume a similar volume for Q1 and Q2 of 2025. Assuming two keyword searches per visit yields approximately 13.8 billion keyword searches per quarter.

Combining these two figures implies an average purchase incidence of

$$\frac{1.2 \text{ billion}}{13.8 \text{ billion}} \approx 0.083.$$

We therefore calibrate each keyword–day market size as

$$M_t = \frac{1}{0.083} \sum_{j \in J} \kappa_{jt} q_{jt}. \quad (\text{A.2})$$

Intuitively, this formula scales observed sales in the focal market up by the inverse of the inside share. The resulting M_t can be interpreted as the number of consumers who search the keyword on day t , including both buyers and non-buyers. The calibration is necessarily approximate, but it provides a transparent mapping from observed keyword-level sales into a market-size measure suitable for the logit demand system.

A.13.3 First-Stage

Results

This subsection reports the first-stage relationship between the price instrument and observed prices. The instrument interacts exchange-rate movements, trade policy uncertainty, and the number of China-origin products in the keyword market. The identifying idea is that these interactions shift seller costs differentially across markets with different exposure to China-origin products, while being plausibly orthogonal to within-market unobserved demand after controlling

for product and keyword fixed effects.

The table below reports the first-stage estimates separately for the pre- and post-tariff periods. The coefficients are positive and precisely estimated in both periods, and the corresponding first-stage F -statistics indicate that the instrument is informative for price.

Table A.9. First Stage Estimates

	Price (1)
$Post=0 \times IV$	0.00056*** (0.00009)
$Post=1 \times IV$	0.00027*** (0.00009)
F -Statistics	28.74 / 28.09
Controls	Yes
ASIN FE	Yes
Keyword FE	Yes
Observations	16,046

Notes: This table reports the first stage results. Standard errors are clustered at the ASIN level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The next table reports the corresponding OLS and IV demand estimates. Reporting both sets of estimates is useful because it illustrates the direction of endogeneity bias: the OLS coefficients imply substantially weaker price sensitivity than the IV specification. The comparison therefore reinforces the importance of instrumenting for price in the structural demand estimation.

Table A.10. Demand Parameter Estimates

	OLS (1)	IV (2)
Price × Pre	-0.033*** (0.007)	-0.169*** (0.042)
Price × Post	-0.033*** (0.006)	-0.143*** (0.041)
Best Seller × Pre	0.801*** (0.116)	0.688*** (0.129)
Best Seller × Post	0.725*** (0.167)	0.812*** (0.171)
Overall Pick × Pre	0.663*** (0.179)	0.531*** (0.170)
Overall Pick × Post	0.810*** (0.134)	0.869*** (0.146)
Rating × Pre	0.462 (0.434)	1.106* (0.571)
Rating × Post	0.535 (0.434)	1.006* (0.572)
Comments × Pre	0.000** (0.000)	0.000** (0.000)
Comments × Post	0.000* (0.000)	0.000* (0.000)
Position × Pre	-0.012*** (0.002)	-0.008*** (0.002)
Position × Post	-0.011*** (0.002)	-0.010*** (0.003)
Observations	16,377	16,046

Notes: This table reports the demand parameters estimates. Standard errors are clustered at the ASIN level and reported in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

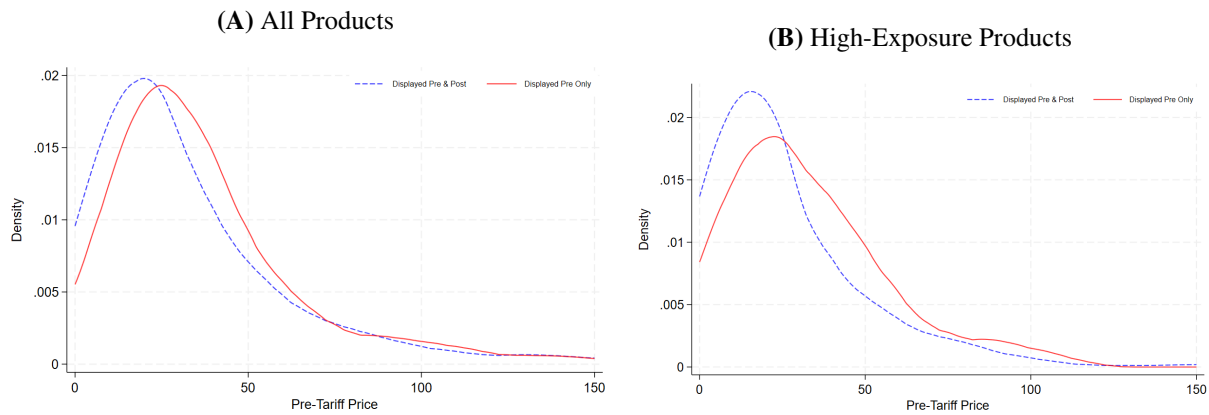
A.14 Pre-Tariff

Price

Distribution

This section provides additional evidence on selection into post-tariff survival. The figure compares the pre-tariff price distribution of products that remain visible in both periods with that of products that appear in the pre period but not in the post period. The purpose is to show that products that later disappear are already more expensive before the tariff shock, which is important for interpreting the seller-side marginal-cost results in the main text.

Figure A.12. Pre-Tariff Price Distribution



Notes: This figure compares the distribution of pre-tariff prices for high-exposure products that remain displayed in both the pre- and post-tariff periods and products that are displayed only in the pre-tariff period.

A.15 Seller

Profit

Decomposition

This section reports the seller-side analogue of the consumer-surplus decomposition in the main text. The purpose is to quantify how the tariff shock affects sellers through the same sequence of channels used in the main welfare analysis: demand shifts, product turnover, changes in product attributes, and platform allocation. As in the main text, the turnover channel is implemented sequentially through separate entry and exit terms.

Average seller profits per consumer increase only slightly, from \$0.506 before the tariff to \$0.510 after the tariff, a gain of \$0.004, or 0.8 percent. As with consumer surplus, this modest net change masks large offsetting components. Evaluated at pre-tariff demand parameters, the post-tariff market would generate seller profits per consumer of \$0.529, so changes in estimated demand reduce seller profits by \$0.018. Entry raises seller profits: moving from the survivor-only post-tariff market to the full post-tariff market increases seller profits per consumer from \$0.498 to \$0.529, a gain of \$0.031. Restoring pre-tariff product attributes for surviving products raises seller profits per consumer from \$0.498 to \$0.604, implying that observed post-tariff changes in prices and other product characteristics reduce seller profits by \$0.106. Restoring the pre-tariff ranking environment further raises seller profits per consumer from \$0.604 to \$0.656, so the post-tariff ranking environment lowers seller profits by an additional \$0.052. Products that later exit account for \$0.150 of pre-tariff seller profits per consumer, or 29.5 percent of the pre-tariff level.

Taken together, these results mirror the consumer-side decomposition. Product turnover partly offsets the negative effects of weaker product attributes and a less favorable allocation of visibility, but the net effect on seller profits remains modest because these forces largely cancel out.

Table A.11. Seller Profit Decomposition

	Initial Level (\$)	Final Level (\$)	Change Level (\$)	Change Percent (%)
	(1)	(2)	(3)	(4)
$\Delta PS: Total$	0.506	0.510	0.004	0.830%
$\Delta PS: Preference$	0.529	0.510	-0.018	-3.461%
$\Delta PS: Entry$	0.498	0.529	0.031	6.245%
$\Delta PS: Attribute$	0.604	0.498	-0.106	-17.583%
$\Delta PS: Algorithm$	0.656	0.604	-0.052	-7.916%
$\Delta PS: Exit$	0.506	0.656	0.150	29.545%

Notes: This table reports average seller profits per consumer at the keyword-generation level. Seller profits per consumer are computed as total variable profits divided by market size. For the total row, columns (1) and (2) report observed pre- and post-tariff levels. For the decomposition rows, columns (1) and (2) report the two objects compared at each step of the sequential decomposition. The preference effect compares post-tariff outcomes under post- and pre-tariff preferences. The entry effect captures post-tariff entrants. The attribute effect restores pre-tariff product attributes for surviving products. The platform-allocation effect restores the pre-tariff ranking rule conditional on pre-tariff attributes. The exit effect compares the rolled-back post-tariff survivor market to the full pre-tariff market. Percent changes are computed relative to column (1).